

14%

SIMILARITY INDEX

PRIMARY SOURCES

- | | | |
|---|--|-----------------|
| 1 | arxiv.org
Internet | 373 words — 2% |
| 2 | www.coursehero.com
Internet | 181 words — 1% |
| 3 | sameni.info
Internet | 79 words — < 1% |
| 4 | documentsdelivered.com
Internet | 66 words — < 1% |
| 5 | Malaya Kumar Nath, Sanghamitra Subhadarsini Dash, Karthick Mookkandi, Thivya A.. "Towards intelligent phonocardiogram analysis: Advances and future perspectives in cardiovascular disease detection", Engineering Applications of Artificial Intelligence, 2026
Crossref | 63 words — < 1% |
| 6 | Shams Nafisa Ali, Afia Zahin, Samiul Based Shuvo, Nusrat Binta Nizam et al. "BUET multi-disease heart sound dataset: A comprehensive auscultation dataset for developing computer-aided diagnostic systems", Computer Methods and Programs in Biomedicine Update, 2026
Crossref | 52 words — < 1% |
| 7 | Yang Yi Poh, Ethan Grooby, Kenneth Tan, Atul Malhotra, Mehrtash Harandi, Faezeh Marzbanrad. "Reprogramming Automatic Speech Recognition | 52 words — < 1% |

Models for Neonatal Chest Sound Separation", IEEE Journal of Biomedical and Health Informatics, 2026

Crossref

8	research.monash.edu Internet	48 words — < 1%
9	Subhasikta Behera, Puneet Kumar Jain, Sambit Bakshi. "Adaptive thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds", Biomedical Signal Processing and Control, 2026 Crossref	47 words — < 1%
10	acikerisim.harran.edu.tr:8080 Internet	45 words — < 1%
11	digitalcommons.odu.edu Internet	44 words — < 1%
12	pubmed.ncbi.nlm.nih.gov Internet	44 words — < 1%
13	Leiming Zhang, Fuliang He, Hao Tan, Wangquan Wang, Shiyuan Wang. "An efficient lung sound separation algorithm base on GIHO-VMD", Computer Methods and Programs in Biomedicine, 2026 Crossref	43 words — < 1%
14	ouci.dntb.gov.ua Internet	42 words — < 1%
15	etd.repository.ugm.ac.id Internet	41 words — < 1%
16	Jaspreet Kaur, Brijesh Kumar. "Comparative Analysis of Machine Learning Algorithms for Lung Sound Analysis for Disease Prediction", 2025 3rd International Conference on Advances in Computation, Communication and Information Technology (ICAICCIT), 2025 Crossref	40 words — < 1%

17	bura.brunel.ac.uk Internet	40 words — < 1%
18	vbn.aau.dk Internet	40 words — < 1%
19	Jing Lyu, Sijie Su, Xin Wang, Enhai Song, Qinyuan Zhang. " High Performance Flexible Optical Fiber Sensor Enabled by Er /Yb Co-Doped near Pure-Green Upconversion Emission Phosphor for Implantable Temperature Sensing ", Advanced Materials Interfaces, 2026 Crossref	38 words — < 1%
20	research.buaa.edu.cn Internet	37 words — < 1%
21	data.mendeley.com Internet	35 words — < 1%
22	Wenhui Sun, Yipeng Zhang, Fuming Chen. "Research on heart and lung sound separation method based on DAE-NMF-VMD", EURASIP Journal on Advances in Signal Processing, 2024 Crossref	34 words — < 1%
23	Daisuke Niizumi, Daiki Takeuchi, Masahiro Yasuda, Binh Thien Nguyen, Yasunori Ohishi, Noboru Harada. "Assessing the Utility of Audio Foundation Models for Heart and Respiratory Sound Analysis", 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2025 Crossref	33 words — < 1%
24	Maros Jakubec, Eva Lieskovska, Peter Pocta. "A robust deep learning based model for denoising phonocardiogram signals in clinical environments", Engineering Applications of Artificial Intelligence, 2025 Crossref	32 words — < 1%

-
- 25 hdl.handle.net 32 words — < 1%
Internet
-
- 26 www.mdpi.com 31 words — < 1%
Internet
-
- 27 Francis Chikweto, Muyanga Kampekete, Mubanga Boyd, Kalupa Conix, Munsanje Foster, Chilufya Queen. "Toward the Development of a Low-Cost AI-Enabled Digital Stethoscope", 2024 International Conference on Electrical and Computer Engineering Researches (ICECER), 2024 30 words — < 1%
Crossref
-
- 28 Peng-Fei Zhang, Hua-Nong Ting, Siow-Wee Chang. "Using Cough Sounds for the Recognition of Multiclass Respiratory Diseases With Artificial Intelligence: A Review", IEEE Access, 2025 30 words — < 1%
Crossref
-
- 29 Van-Thuan Tran, Ting-Hao You, Wei-Ho Tsai. "Acoustic Cues for Person Identification Using Cough Sounds", Computer Methods and Programs in Biomedicine Update, 2025 30 words — < 1%
Crossref
-
- 30 Alessandra Fava, Behnood Dianat, Alessandro Bertacchini, Andreina Manfredi, Marco Sebastiani, Marco Modena, Fabrizio Pancaldi. "Pre-processing techniques to enhance the classification of lung sounds based on deep learning", Biomedical Signal Processing and Control, 2024 29 words — < 1%
Crossref
-
- 31 core.ac.uk 29 words — < 1%
Internet
-
- 32 sciendo.com 29 words — < 1%
Internet

-
- 33 open.library.ubc.ca 27 words — < 1%
Internet
-
- 34 Ming Hung Wong. "Handbook of Microplastics and Associated Chemicals - Ecological and Health Impacts", CRC Press, 2026 25 words — < 1%
Publications
-
- 35 Xu Wang, Hui Zhou, Yanjie Xu, Yingtao Zhang, Zhenguo Nie. "Multimodal Acoustic Feature Fusion with Hybrid Deep Networks for Industrial Equipment Fault Detection", 2025 International Conference on Equipment Intelligent Operation and Maintenance (ICEIOM), 2025 25 words — < 1%
Crossref
-
- 36 nitrkl.ac.in 25 words — < 1%
Internet
-
- 37 Chenwen Wu, Na Ye, Jialin Jiang. "Classification and Recognition of Lung Sounds Based on Improved Bi-ResNet Model", IEEE Access, 2024 24 words — < 1%
Crossref
-
- 38 www.db-thueringen.de 24 words — < 1%
Internet
-
- 39 www.medrxiv.org 24 words — < 1%
Internet
-
- 40 doaj.org 23 words — < 1%
Internet
-
- 41 Bowen Han, Wei Quan. "Cardiorespiratory Sound Separation Using Singular Spectrum Analysis", 2025 17th International Conference on Signal Processing Systems (ICSPS), 2025 22 words — < 1%
Crossref

42 Rizwan Ullah, Shaohui Zhang. "Unsupervised framework for single channel heart and lung sounds separation in data constrained environments", *Applied Acoustics*, 2024
Crossref

22 words — < 1%

43 eprints.utar.edu.my
Internet

22 words — < 1%

44 iieta.org
Internet

22 words — < 1%

45 "Table of Contents", *IEEE Journal of Biomedical and Health Informatics*, 2026
Crossref

19 words — < 1%

46 Wangquan Wang, Chenghong Bo, An Zhao, Leiming Zhang, Fuliang He. "Heart-lung sound unsupervised separation by nonnegative matrix factorization and modified Hilbert spectrum", *Biomedical Signal Processing and Control*, 2025
Crossref

19 words — < 1%

47 assets-eu.researchsquare.com
Internet

19 words — < 1%

48 www.newbornjournal.org
Internet

19 words — < 1%

49 suraj.lums.edu.pk
Internet

18 words — < 1%

50 Bochao Zhang, Jiping Wang, Zhipeng Ye, Linfu Zhou, Daxi Xiong, Xiaojun Wang, Liquan Guo. "JASSNet: Heart and lung sound separation network based on joint attention mechanism and Semi-Supervised learning", *Biomedical Signal Processing and Control*, 2025
Crossref

17 words — < 1%

51 Yanshan Sha, Dongliang Chen, Feiyang Wu, Qiang Cao, Yucong Zhou, Weihua Zhang, Xin Huang, Minghui Wang. "Experimental investigation of time-frequency characteristics of acoustic signals during boilover in small-scale oil tanks", Journal of Loss Prevention in the Process Industries, 2026

Crossref

17 words — < 1%

52 equip.sabah.gov.my

Internet

17 words — < 1%

53 erepo.uef.fi

Internet

17 words — < 1%

54 jeeemi.org

Internet

17 words — < 1%

55 H.L. Gururaj, Francesco Flammini, S. Srividhya, M.L. Chayadevi, Sheba Selvam. "Computer Science Engineering", CRC Press, 2024

Publications

16 words — < 1%

56 Yuqi Wang, Han Yang, Zhixing Gao, Zhiwei Dai, Kang Yu, Tingting Song, Hao Yang, Yunfeng Wang, Jun Zhang. "A Phase-Enhanced Neural Network With Dual-Path Transformer for Single-Channel Chest Sound Separation", IEEE Journal of Biomedical and Health Informatics, 2026

Crossref

16 words — < 1%

57 digibuo.uniovi.es

Internet

16 words — < 1%

58 ds.inflibnet.ac.in

Internet

16 words — < 1%

59 eprints.staffs.ac.uk

Internet

16 words — < 1%

-
- 60 "Swarm, Evolutionary, and Memetic Computing", Springer Science and Business Media LLC, 2015
Crossref 15 words — < 1%
-
- 61 Ayman Anwar, Yassin Khalifa, James L. Coyle, Ervin Sejdic. "Transformers in biosignal analysis: A review", Information Fusion, 2025
Crossref 15 words — < 1%
-
- 62 Ravi Pal, Anna Barney. "Automatic breathing phase identification based on the second derivative of the recorded lung sounds", Biomedical Signal Processing and Control, 2024
Crossref 15 words — < 1%
-
- 63 [github.com](#)
Internet 15 words — < 1%
-
- 64 [pdfs.semanticscholar.org](#)
Internet 15 words — < 1%
-
- 65 "2023 Index IEEE Journal of Biomedical and Health Informatics Vol. 27", IEEE Journal of Biomedical and Health Informatics, 2023
Crossref 14 words — < 1%
-
- 66 [oro.open.ac.uk](#)
Internet 14 words — < 1%
-
- 67 [www.papers.phmsociety.org](#)
Internet 14 words — < 1%
-
- 68 "Applications of Computational Intelligence in Management & Mathematics", Springer Science and Business Media LLC, 2023
Crossref 13 words — < 1%
-
- 69 E. Grooby, C. Sitaula, D. Fattahi, R. Sameni, K. Tan, L. Zhou, A. King, A. Ramanathan, A. Malhotra, G.A. Dumont, F. Marzbanrad. "Noisy Neonatal Chest Sound 13 words — < 1%

-
- 70 project-archive.inf.ed.ac.uk 13 words — < 1%
Internet
-
- 71 researchr.org 13 words — < 1%
Internet
-
- 72 www.peeref.com 13 words — < 1%
Internet
-
- 73 Lok, Richard Him. "Tool Support for Software Process Assessments", University of the Witwatersrand, Johannesburg (South Africa), 2025 12 words — < 1%
ProQuest
-
- 74 Weibo Wang, Shubo Wang, Dimei Qin, Yu Fang, Yongkang Zheng. "Heart-lung sound separation by nonnegative matrix factorization and deep learning", Biomedical Signal Processing and Control, 2023 12 words — < 1%
Crossref
-
- 75 Yasaman Torabi, Shahram Shirani, James P. Reilly. "LingoNMF: Large language model-enhanced nonnegative matrix factorization for cardiorespiratory sound separation", Biomedical Signal Processing and Control, 2026 12 words — < 1%
Crossref
-
- 76 assets.researchsquare.com 12 words — < 1%
Internet
-
- 77 online-journals.org 12 words — < 1%
Internet
-
- 78 portalcris.lsmuni.it 12 words — < 1%
Internet
-
- 79 scholarshare.temple.edu

12 words — < 1%

80 "Table of Contents", IEEE Sensors Journal, 2025 11 words — < 1%
Crossref

81 Ethan Grooby, Jinyuan He, Julie Kiewsky, Davood Fattahi et al. "Neonatal Heart and Lung Sound Quality Assessment for Robust Heart and Breathing Rate Estimation for telehealth Applications", IEEE Journal of Biomedical and Health Informatics, 2020 11 words — < 1%
Crossref

82 kipdf.com 11 words — < 1%
Internet

83 mobt3ath.com 11 words — < 1%
Internet

84 M. Gayathri, Jogendra Kumar M, Chandu Jagan Sekhar Madala, Shanmuk Srinivas Amiripalli et al. "A self-Supervised ConvNeXt (SCONE) framework for robust and interpretable heart sound classification", Intelligence-Based Medicine, 2026 10 words — < 1%
Crossref

85 Osazuwa, Lovelina I.. "Comparison of Christian Television Networks in the United States and Nigeria (2005–2025)", Drexel University, 2025 10 words — < 1%
ProQuest

86 Reshma Sreejith, R. Kanesaraj Ramasamy, Wan-Noorshahida Mohd-Isa, Junaidi Abdullah. "Review of advancements in AI-assisted lung sound analysis for respiratory illness diagnosis in noisy environments", IAES International Journal of Artificial Intelligence (IJ-AI), 2026 10 words — < 1%
Crossref

87 theses.whiterose.ac.uk 10 words — < 1%
Internet

88	ijcrt.org Internet	10 words — < 1%
89	ijmada.com Internet	10 words — < 1%
90	os.unil.cloud.switch.ch Internet	10 words — < 1%
91	researchbank.swinburne.edu.au Internet	10 words — < 1%
92	scholar.hit.edu.cn Internet	10 words — < 1%
93	"Energy-Efficient Algorithms and Systems in Computing", Springer Science and Business Media LLC, 2026 Crossref	9 words — < 1%
94	Lemos, Diogo Azevedo. "AI-Based Tool for Lifecycle Management and Preservation of Architectural Questions.", Universidade do Porto (Portugal) ProQuest	9 words — < 1%
95	ase.in.tum.de Internet	9 words — < 1%
96	baltijapublishing.lv Internet	9 words — < 1%
97	cdn.techscience.cn Internet	9 words — < 1%
98	ebin.pub Internet	9 words — < 1%
99	researchbank.rmit.edu.au Internet	9 words — < 1%

-
- 100 www.researchsquare.com 9 words — < 1%
Internet
-
- 101 Ali Hussein Shaker, Ibrahim Amer Ibrahim, Sadik Kamel Gharghan. "Machine learning techniques for cardiovascular disease detection through heart sound analysis: A review", AIP Publishing, 2024 8 words — < 1%
Crossref
-
- 102 Green, Virginia. "Entrepreneurial boot camps: A case study to evaluate and improve the establishment of methods, goals, strategies and tasks", Proquest, 20111109 8 words — < 1%
ProQuest
-
- 103 K. Sathesh, P Kowsalya, E. Aravindraj, S.Guru Naveen Reddy, K. Anvith Sai Varma, P. Ganesh. "Clearer HeartBeats: Enhancement of cardiac sounds using Adaptive Filtering and Wavelet Decomposition", 2024 10th International Conference on Communication and Signal Processing (ICCSP), 2024 8 words — < 1%
Crossref
-
- 104 R. N. V. Jagan Mohan, B. H. V. S. Rama Krishnam Raju, V. Chandra Sekhar, T. V. K. P. Prasad. "Algorithms in Advanced Artificial Intelligence - Proceedings of International Conference on Algorithms in Advanced Artificial Intelligence (ICAAl-2024)", CRC Press, 2025 8 words — < 1%
Publications
-
- 105 Shuqi Ren, Wei Zhao, Zengsheng Chen, Li Wang, Yuheng Yang, Xiaoyan Deng, Yubo Fan, Anqiang Sun. "New insights into the 120-year-old Korotkoff sound, which would reflect artery stiffness and more cardiovascular diseases", Results in Engineering, 2026 8 words — < 1%
Crossref
-
- 106 Yuan Xie, Kan Xie, Qiyu Yang, Shengli Xie. "Reverberant blind separation of heart and lung 8 words — < 1%

sounds using nonnegative matrix factorization and auxiliary function technique", Biomedical Signal Processing and Control, 2021

Crossref

107	ediss.sub.uni-hamburg.de Internet	8 words — < 1%
108	eprints.bournemouth.ac.uk Internet	8 words — < 1%
109	es.scribd.com Internet	8 words — < 1%
110	etd.uum.edu.my Internet	8 words — < 1%
111	journal.50sea.com Internet	8 words — < 1%
112	journals.plos.org Internet	8 words — < 1%
113	knowledge.lancashire.ac.uk Internet	8 words — < 1%
114	par.nsf.gov Internet	8 words — < 1%
115	studentsrepo.um.edu.my Internet	8 words — < 1%
116	www.csauthors.net Internet	8 words — < 1%
117	Adekanmbi, Motunrayo Helen. "Knowledge of Tuberculosis Patients About Their Disease in Tshwane, South Africa", University of South Africa (South Africa) ProQuest	7 words — < 1%

118 Chunjian Yang, Nan Hu, Dongyang Xu, Zhi Wang, Shengsheng Cai. "Monaural cardiopulmonary sound separation via complex-valued deep autoencoder and cyclostationarity", Biomedical Physics & Engineering Express, 2023

7 words — < 1%

Crossref

119 Declan Duggan, Andriy Temko, Volodymyr Sarana, Andreea Factor, Emanuel Popovici. "Denoising of Heart Sounds Using Lightweight FCNs and Spectrograms With and Without Context", IEEE Access, 2025

7 words — < 1%

Crossref

120 Eray Mutlu, Valid Hüseyin, Görkem Serbes. "Bridging Auscultation and Tiny Machine Learning: A Digital Stethoscope Leveraging Convolutional Neural Networks on an Embedded Device for Organ Sound Analysis", Traitement du Signal, 2024

7 words — < 1%

Crossref

121 Gopinath Rajarathinam, Radhamani Rajakannan, Rajan Chinnasamy, Kavinesh Karthikeyan et al. "A Hybrid CNN-BiLSTM Model for Real-Time Activity Classification in Distributed Acoustic Sensing Systems", Traitement du Signal, 2026

7 words — < 1%

Crossref

122 Liuping Chen, Wensheng Chen, Huiqi Chen, Mingli Guo et al. "Edible Plant-Derived Extracellular Vesicles: Emerging Nanocarriers for Regenerative Medicine", Journal of Future Foods, 2026

7 words — < 1%

Crossref

123 Mahmoud Fakhry, Abeer FathAllah Brery. "Generative artificial intelligence for heart sound analysis: From model architectures to clinical applications", Intelligent Systems with Applications, 2026

7 words — < 1%

Crossref

124

Shuvashis Sarker, Faika Fairuj Preotee, Tashreef Muhammad, Shamim Akhter. "An Innovative Framework for Heart Sound Classification Integrating Adaptive Fuzzy Rank-Based Ensemble of Transfer Learning Models with Multi-Dimensional Features", Cold Spring Harbor Laboratory, 2025

Crossref Posted Content

7 words — < 1%

125

Chiranji Lal Chowdhary, G. Thippa Reddy, B. D. Parameshachari. "Computer Vision and Recognition Systems - Research Innovations and Trends", Routledge, 2022

Publications

6 words — < 1%

126

Ganesh R. Naik. "Time-Frequency Analysis in Biomedical Engineering - Contemporary Methods and Applications", CRC Press, 2026

Publications

6 words — < 1%

127

Meagan Roff, Olivia Slifirski, Ethan Grooby, Faezeh Marzbanrad, Atul Malhotra. "Digital Stethoscope Use in Neonates: A Systematic Review", Newborn, 2023

Crossref

6 words — < 1%



FACULTY OF
COMPUTING
& INFORMATICS

FINAL YEAR PROJECT INTERIM REPORT

985

Machine Learning-Based System for Cardiopulmonary Sound
Separation

1221303805

AL-SALOUL, ASHRAF ALI HUSSEIN

²
BACHELOR OF COMPUTER SCIENCE B.CS (HONS) SOFTWARE
ENGINEERING

JULY 2026

985
Machine Learning-Based System for Cardiopulmonary
Sound Separation

BY

1221303805 AL-SALOUL, ASHRAF ALI HUSSEIN

PROJECT INTERIM REPORT ⁵² SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENT FOR THE DEGREE OF
BACHELOR OF COMPUTER SCIENCE B.CS (HONS) SOFTWARE ENGINEERING

⁸⁶
in the

Faculty of Computing and Informatics

MULTIMEDIA UNIVERSITY
MALAYSIA

July 2026

Copyright

© 2026 ²Universiti Telekom Sdn. Bhd. ALL RIGHTS RESERVED.

Copyright of this report belongs to Universiti Telekom Sdn. Bhd. as qualified by Regulation 7.2 (c) of the Multimedia University Intellectual Property and Commercialisation Policy. No part of this publication may be reproduced, stored in or introduced into a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), or for any purpose, without the express written permission of Universiti Telekom Sdn. Bhd. Due acknowledgement shall always be made of the use of any material contained in, or derived from, this report.

Declaration

I hereby declare that the work has been done by myself and no portion of the work contained in this report has been submitted in support of any application for any other degree or qualification of this or any other university or institution of learning.



Name of candidate: AL-SALOUL, ASHRAF ALI HUSSEIN

Faculty of Computing & Informatics

Multimedia University

Date: 05: 07: 2026

Acknowledgements

In the name of Allah, the Most Gracious, the Most Merciful.

I would like to express my sincere gratitude to my supervisor, Arshad Jamal, for his guidance, patience, and continuous support throughout the development of this project. His advice has been valuable in shaping both the technical and academic direction of this work.

I am also grateful to Multimedia University and the Faculty of Computing and Informatics for providing the academic environment, resources, and support necessary to complete this project.

Finally, I would like to thank my family, friends, and colleagues for their encouragement, patience, and moral support throughout this journey.

Abstract

A cardiopulmonary sound recording may contain heart sounds, lung sounds, sensor contact noise, and environmental artefacts in one audio signal. Because these sounds may overlap, using only a simple filter may not produce clear heart and lung outputs. The proposed project is titled **Machine Learning-Based System for Cardiopulmonary Sound Separation**, and its main objective is to design a software prototype that can accept mixed cardiopulmonary audio, perform suitable preprocessing, and generate separate audio outputs for heart and lung sounds. The proposed study also plans to plan the evaluation of the designed software prototype using suitable datasets and metrics for cardiopulmonary sound separation.

Based on an application-based prototype approach, the report is organized accordingly. In the literature review, the sound characteristics of the cardiopulmonary system are discussed, along with machine learning and deep learning methods, available cardiopulmonary datasets, evaluation metrics, and gaps in software prototypes. The report also explains the requirements analysis and system design before describing the planned implementation workflow. This workflow covers audio upload, validation, preprocessing, model selection, separation, output handling, and evaluation planning. The project is not geared towards disease diagnosis or clinical decision-making, but rather provides a structured software workflow for research and experimentation in the area of cardiopulmonary sound separation.

Table of Contents

82	Copyright	iii
	Declaration	iv
	Acknowledgements.....	v
	Abstract	vi
	Table of Contents.....	vii
	List of Tables.....	x
	List of Figures	xi
	List of Abbreviations/Symbols.....	xii
	List of Appendices	xiii
	CHAPTER 1: INTRODUCTION	1
	1.1 Overview	1
	1.2 Problem Statement	1
	1.3 Project Objectives	2
	1.4 Project Scope.....	3
	1.5 Project Limitations.....	3
	1.6 Methodology.....	3
	1.7 Target Audience	4
	1.8 Summary.....	4
	CHAPTER 2: LITERATURE REVIEW	5
	2.1 Overview	5
	2.2 Literature Search and Screening Process.....	5
	2.3 Cardiopulmonary Sound Characteristics.....	6
	2.4 Heart and Lung Sound Separation Techniques.....	7
	2.4.1 Traditional Signal Processing Methods.....	7
	2.4.2 Machine Learning and Deep Learning Methods.....	7
	2.5 Datasets for Cardiopulmonary Sound Separation.....	8
	2.6 Evaluation Metrics.....	8
	2.7 Existing Systems and Prototype Relevance.....	9
	2.8 Literature Review Matrix	9
	2.9 Research Gaps	10

2.10 Summary.....	49 10
CHAPTER 3: REQUIREMENTS ANALYSIS.....	12
3.1 Overview.....	12
3.2 Questionnaire.....	12
3.2.1 Questionnaire Design	12
3.2.2 Respondent Background and Experience	14
3.2.3 Question 1: Difficulty of Separating Heart and Lung Sounds.....	15
3.2.4 Question 2: Problems Affecting Cardiopulmonary Sound Processing	16
3.2.5 Question 3: Usefulness of a Heart and Lung Separation Prototype .	17
3.2.6 Question 4: Preferred System Features	18
3.2.7 Question 5: Preferred Output after Sound Separation.....	19
3.2.8 Question 6: Importance of Audio Preview Function	20
3.2.9 Question 7: Preferred Interface Style	21
3.2.10 Question 8: First Information Shown after Separation	22
3.2.11 Question 9: Importance of Evaluation Metrics	23
3.2.12 Question 10: Willingness to Use the Prototype.....	24
3.3 Functional Requirements	26
3.4 Non-Functional Requirements	26
3.5 User Requirements	43 27
3.6 Summary.....	28
CHAPTER 4: SYSTEM DESIGN	29
4.1 Overview.....	29
4.2 Context Diagram	29
4.3 Use Case Diagram.....	30
4.3.1 Use Case Description	31
4.4 Activity Diagram	34
4.5 Class Diagram.....	35
4.5.1 Component-Level Architecture	37
4.5.2 Algorithm and Model Selection Design	38
4.5.3 Database and Storage Design.....	39
4.6 Sequence Diagram	40

4.7 Interface Design	41
4.8 Summary.....	42
CHAPTER 5: IMPLEMENTATION PLAN	43
5.1 Development Phase	43
5.1.1 Front-End Development	43
5.1.2 Back-End Development.....	44
5.1.3 Machine Learning Model Integration	44
5.1.4 Database and Storage Integration.....	49
5.2 Testing	49
5.2.1 Unit Testing.....	50
5.2.2 Integration Testing	50
5.2.3 System Testing	50
5.2.4 User Acceptance Testing	51
5.3 Deployment.....	51
5.4 Documentation and Maintenance Plan	52
5.5 Summary.....	52
CHAPTER 6: CONCLUSION.....	53
6.1 Overview	53
6.2 Project Summary.....	53
6.3 Expected Contributions	53
6.4 Limitations.....	53
6.5 Future Work	54
6.6 Summary.....	54
References	xiv
Appendix A: Gantt Chart	xviii
Appendix B: FYP1 Meeting Logs.....	xix
Appendix C: Turnitin Similarity Index Page.....	xxi
Appendix D: Full Literature Review Matrix.....	xxvi
Appendix E: PRISMA Screening Summary	xxxiii
Appendix F: System Design Diagrams	xxxiv
Appendix G: Planned Test Cases and Evaluation Metrics	xxxv
Appendix H: Full User Requirements Questionnaire	xxxvi

List of Tables

Table 1: Summary literature review matrix.....	9
Table 2: Questionnaire structure for the user requirements analysis	13
Table 3: Questionnaire analysis mapping	14
Table 4: Functional requirements for cardiopulmonary sound separation prototype.....	26
Table 5: Non-functional requirements for cardiopulmonary sound separation prototype.....	27
Table 6: User requirements for proposed prototype.....	27
Table 7: Major use case description for the cardiopulmonary sound separation prototype.....	32
Table 8: Main system modules and responsibilities	37
Table 9: Options for model selection in the cardiopulmonary sound separation prototype.....	39
Table 10: Database entities for prototype traceability and analysis	40
Table 11: Interface areas for cardiopulmonary sound separation	42
Table 12: Planned backend API responsibilities	44

List of Figures

Figure 1: PRISMA flow diagram.....	6
Figure 2: Perceived difficulty of separating heart and lung sounds.....	16
Figure 3: Problems affecting cardiopulmonary sound processing	17
Figure 4: Usefulness of the proposed heart and lung separation prototype	18
Figure 5: Preferred system features	19
Figure 6: Preferred output after sound separation	20
Figure 7: Preview separated heart and lung sounds	21
Figure 8: Preferred interface style	22
Figure 9: Preferred first information after separation is completed	23
Figure 10: Importance of evaluation metrics for separated output.....	24
Figure 11: Willingness to use the cardiopulmonary sound separation prototype	25
Figure 12: Context diagram for the proposed cardiopulmonary sound separation prototype.....	30
Figure 13: Use case diagram for the cardiopulmonary sound separation prototype.....	31
Figure 14: Activity diagram for the sound separation workflow	35
Figure 15: Class diagram for the cardiopulmonary sound separation prototype	36
Figure 16: Component-level architecture for the cardiopulmonary sound separation prototype	37
Figure 17: Sequence diagram for uploading and separating a cardiopulmonary recording.....	41

List of Abbreviations/Symbols

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DL	Deep Learning
EPS	Small epsilon value used to avoid division by zero
FFT	Fast Fourier Transform
FYP	Final Year Project
HLS-CMDS	Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope
HTTP	Hypertext Transfer Protocol
ID	Identifier
ML	Machine Learning
NeoSSNet	Deep learning model for neonatal chest sound separation
NMF	Non-negative Matrix Factorization
PCG	Phonocardiogram
PESQ	Perceptual Evaluation of Speech Quality
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
S1	First heart sound
S2	Second heart sound
SAR	Signal-to-Artifacts Ratio
SDR	Signal-to-Distortion Ratio
SIR	Signal-to-Interference Ratio
SI-SDR	Scale-Invariant Signal-to-Distortion Ratio
SNR	Signal-to-Noise Ratio
SQLite	Lightweight relational database engine
STFT	Short-Time Fourier Transform
STOI	Short-Time Objective Intelligibility
UI	User Interface
UML	Unified Modeling Language
VMD	Variational Mode Decomposition
WAV	Waveform Audio File Format

List of Appendices

<u>Appendix A: Gantt Chart</u>	xviii
<u>Appendix B: FYP1 Meeting Logs</u>	xix
<u>Appendix C: Turnitin Similarity Index Page</u>	xxi
<u>Appendix D: Full Literature Review Matrix</u>	xxvi
<u>Appendix E: PRISMA Screening Summary</u>	xxxiii
<u>Appendix F: System Design Diagrams</u>	xxxiv
<u>Appendix G: Planned Test Cases and Evaluation Metrics</u>	xxxv
<u>Appendix H: Full User Requirements Questionnaire</u>	xxxvi

CHAPTER 1: INTRODUCTION

1.1 Overview

Cardiopulmonary auscultation captures sounds produced by the heart, lungs, airways, the chest wall and the recording environment. Typical recordings contain useful information about the cardiovascular and respiratory systems masked by artifacts such as breathing sounds, body movement, motion, sensor noise and room noise. This renders a practical software design challenge since the user is not interested in a mixture but rather another recording without the unwanted contribution. This project will focus on the design of a prototype workflow that can be reused to separate cardiopulmonary sounds.

The project's name is Machine Learning Based System for Cardiopulmonary Sound Separation. This is an application based FYP1 project and the contribution is prototype/proof of concept and NOT a medical product. This stage is to prepare the foundation for the FYP2 which consists of background study, literature review, requirements analysis, system design and implementation plan.

1.2 Problem Statement

1. Noise in a recording, plus additional artefacts generated during the recording, will degrade the quality of your Cardiopulmonary Sound Recordings before even attempting to process them with any software. In practice, sensor noises, artefacts from breathing movements, unwanted movement artefacts, background noises, and limitations of recordable quantities by the recording devices, can contaminate heart and lung sounds. The most important limitation here is that this is a serious concern to all of our separation tasks, because it means that the model will be fed with a distorted signal as input instead of the ideal clean signal. Research works regarding noisy neonatal chest sounds, blind denoising, and lung sound enhancement have demonstrated that preprocessing is an essential step before attempting further downstream analysis or separation ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Ali et al., 2023](#)). This means that for our FYP, the choice of model is not the only consideration. Your software will also require file validation and additional preprocessing before the actual separation can take place.

2. Heart and lung sounds are also not easy to differentiate since they may co-occur both in time and frequency. Short heart sound events and longer respiratory patterns can be simultaneously recorded on a cardiopulmonary recording. These frequency ranges can overlap, and a set frequency filter could filter out information or introduce sound that is not required in the separated output. In the case of a single mixed recording, without separate original heart and lung sources available, the problem is more difficult to solve than the prototype. Recent work on neonatal heart and lung sound separation, decomposition-based methods, and compact learning models suggests that this task needs more than a simple filtering rule (Poh et al., 2024; Sun et al., 2024; Wang et al., 2023a). This project responds to that gap by planning a machine learning-based separation workflow that can support different separation strategies during FYP2.
3. A third difficulty is that there is no clear software re-usable process for students/researchers interested in using cardiopulmonary sound separation as an application. The majority of related studies are diagnosis, classification or algorithm level experiment. The above studies are useful but not always illustrative of the steps a user would take in uploading one mixed recording and validating it, preprocessing it, choosing a model, running separation, and downloading the separated heart and lung outputs. Review and analysis papers also reveal that much of the research in the field of cardiopulmonary sounds focuses on the clinical interpretation or the model performance instead of the surrounding prototype design (Roy et al., 2024; Zhao et al., 2024; Torabi et al., 2025a). Since this is a Software Engineering FYP, the system design matters as much as the selected algorithm. The prototype must be based on the following requirements: clear requirement, responsibilities of modules, flow of the interface, planning of storage and planning of evaluation, so as to be able to be implemented in FYP2 in a controlled manner.

1.3 Project Objectives

1. To identify and apply relevant preprocessing techniques to reduce noise and enhance the quality of the cardiopulmonary recordings.

2. To propose a machine learning-based method that can separate mixed cardiopulmonary recordings into heart and lung sound outputs.
3. To develop a prototype, reusable in research, and evaluate the developed tool using appropriate public datasets and performance metrics.

1.4 Project Scope

The current project scope is a prototype for cardiopulmonary separation software. This prototype aims to receive one input file with a mixed cardiopulmonary recording, verify it, apply preprocessing, execute the selected procedure for separation, and output two audio files for preview and download.

As per the project's scope, this work will involve publicly available or otherwise approved datasets, preprocessing, model integration, and evaluation planning with relevant performance metrics. This prototype is not intended to detect, diagnose, or recommend any medical treatment or procedure. It is also not designed to operate in a clinical environment or replace the professional judgment of a medical practitioner. Instead, it will serve as a research and educational prototype for separation tasks only.

1.5 Project Limitations

The project's scope is limited to the FYP1 stage; thus, the current report will focus on background, requirements, design, and planning. The actual model training, implementation, user acceptance, and performance assessment will be carried out in the FYP2 stage. Similarly, the proposed prototype is limited by the availability of public or otherwise approved cardiopulmonary recordings. The data gathered from the questionnaires are based on the convenience sample of students, and thus they only represent general requirements. This limitation is standard for FYP1, but it is essential to acknowledge it when elaborating on the findings.

1.6 Methodology

This work will follow the Application-Based methodology that involves investigation, definition, design, and planning phases. First, the project will review the characteristics of cardiopulmonary recordings, potential separation procedures, public datasets, and performance metrics. Second, the requirements will be established by analyzing the literature, existing solutions or systems, supervisor's notes, and the requirements

survey results. Third, the design will be created with the support of multiple artefacts, including Context, ⁴³ Use Case, and Activity Diagrams for the process and the Class and Sequence Diagrams for implementation specifics. Finally, the FYP1 will conclude with the planning of Front End, Back End, machine learning integration, storage, testing, deployment, documentation, and maintenance stages in FYP2.

This methodology will help this project to follow the clear logic of the Application-Based task. It will also support the report's claims by separating requirements, design, and implementation phases. In particular, the FYP1 report will detail the findings, but it will avoid presenting results that will only be available in the FYP2 stage.

1.7 Target Audience

The project's target audience is the students, researchers, and evaluators interested in an accessible cardiopulmonary separation workflow. The prototype under development will contribute to these audiences' needs by providing a step-by-step process for separating mixed cardiopulmonary recordings into heart and lung sound outputs.

Furthermore, the project will benefit the student developer by structuring the FYP2 work and outlining the tasks of input file verification, preprocessing, model training, cardiopulmonary separation, storage, preview, download, and performance evaluation.

1.8 Summary

This chapter described the project's background, scope, limitations, methodology, and audience. The project aims to develop a prototype of cardiopulmonary separation software. The primary challenge is to ensure that the cardiopulmonary recordings will remain consistent in terms of quality and free from noise and interference. It was found that many works report cardiopulmonary separation, but the focus of this project is to provide a complete prototype for the separation task. The next chapter will provide details on the relevant literature and research findings.

91 CHAPTER 2: LITERATURE REVIEW

2.1 Overview

The review focuses on the studies that served as the foundation for developing the cardiopulmonary sound separation prototype. Specifically, the review dwells on sound characteristics, separation methods, datasets, evaluation criteria, existing systems, and the relevance of prototypes to cardiopulmonary separation. Thus, the review's objective is to furnish a basis for the cardiopulmonary prototype in terms of data preprocessing, model-based processing, outputs, and evaluation.

The common problem with the prototypes described in the reviewed articles is the presence of cardiac, pulmonary, and recording artifacts in the same channel. The conventional signal processing approaches can partly address this issue, as they served as the foundation for building more complex models. Nevertheless, the current state-of-the-art in machine learning and deep learning seems to be more relevant to the task at hand. The problem of sound separation is more involved than linear signal processing and benefits from the flexibility of neural networks.

2.2 Literature Search and Screening Process

The search strategy comprised cardiopulmonary separation methods, including cardiopulmonary sound separation, heart sound removal algorithms, lung sound separation, and denoising. The strategy identified the relevant literature on public auscultation databases, evaluation metrics, and cardiopulmonary sound prototype systems. Initially, 96 records were retrieved; however, only 49 were included in the final screening process. Consequently, 35 articles were selected for inclusion in this review and the subsequent creation of a literature matrix for the full systematic review. In total, 18 articles were included in the main topic of cardiopulmonary separation methods, and 17 constituted the background/supplementary category. Finally, 14 records were excluded from the current study but are available for future research.

The review follows the PRISMA methodology, and Figure 1 below summarizes the literature search and screening process.

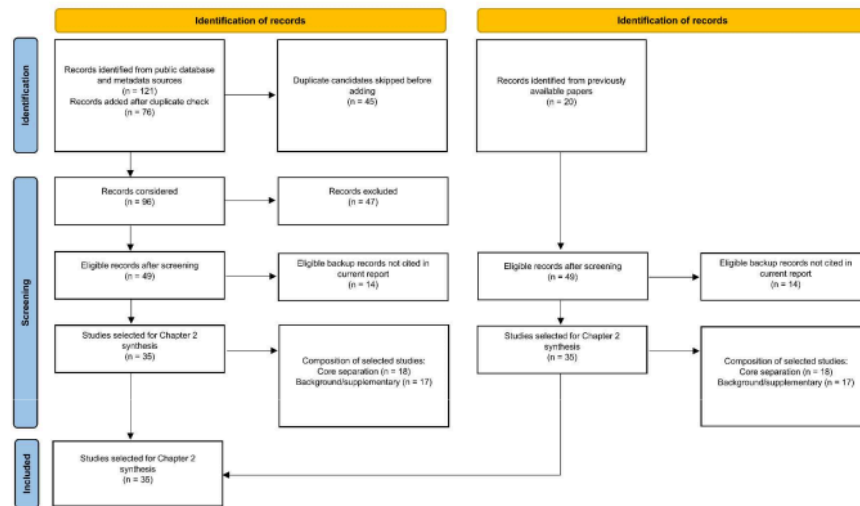


Figure 1: PRISMA flow diagram

2.3 Cardiopulmonary Sound Characteristics

Recordings of cardiopulmonary sounds contain a combination of cardiac, pulmonary, chest wall, and recording artifacts vibrations. Heart activity is typically present in the form of periodically occurring S1 and S2 clicks. Lung sounds are associated with the inhalation-exhalation process and vary depending on the stage of breathing. Since both heart and lung activity are recorded from the same surface, their spectra may overlap, making separation challenging. This effect is particularly noticeable when using recordings from a single channel.

The phenomenon is explained in detail in the studies on respiratory sound analysis and cardiopulmonary sound characteristics (Roy et al., 2024; Zhao et al., 2024; Grooby et al., 2023). According to these records, a cardiopulmonary sound separation prototype must address the interference from multiple sources and varying levels of noise. Hence, the necessity of preprocessing and standardized input preparation for model-based separation.

2.4 Heart and Lung Sound Separation Techniques

Heart and lung sound separation is a well-researched topic that includes traditional signal processing approaches and machine learning methods. In particular, cardiopulmonary separation research shines a light on the significance of selecting separation methods based on specific conditions, such as recording sources, noise levels, training data availability, and evaluation criteria.

2.4.1 Traditional Signal Processing Methods

Traditional signal processing methods include filtering, adaptive filtering, blind source separation, non-negative matrix factorization, empirical mode decomposition, variational mode decomposition, and similar techniques. The primary benefit of such an approach is its explainability, which is critical for a cardiopulmonary sound separation prototype. Moreover, many signal processing methods can be applied without extensive training data. Blind denoising and decomposition-based separation are super relevant to cardiopulmonary separation, as they tackle the issues of mixed heart and lung recordings (Fattahi et al., 2022; Sun et al., 2024; Ullah and Zhang, 2024).

However, the limitations of traditional methods are also considerable, particularly the inability to manage complex or varying sources of interference. Hence, a method that works for one set of conditions may fail for another recording. For these reasons, traditional signal processing methods can be used as a baseline or as part of a prototype's preprocessing stage.

2.4.2 Machine Learning and Deep Learning Methods

Machine learning and deep learning approaches to cardiopulmonary separation learn the patterns that can distinguish between heart and lung sounds. Recent research points out the effectiveness of neural networks, source separation algorithms, compact models, and data-driven methods for cardiopulmonary sound analysis (Poh et al., 2024; Wang et al., 2023a; Zhang et al., 2025; Ali et al., 2023). These methods are super relevant to the current project because they can be trained to process a single input and separate heart and lung sounds.

The main advantage of machine learning approaches is their ability to capture complex patterns in data, including those that cannot be described by fixed rules. The limitation

of such methods is their dependence on training data and the computational power of the underlying hardware. Hence, the cardiopulmonary separation model must be trained on a suitable dataset and have consistent preprocessing and evaluation pipelines. Finally, the choice of a particular algorithm is not critical to the current research since the primary focus is on the prototype as a whole.

2.5 Datasets for Cardiopulmonary Sound Separation

Public datasets are crucial for cardiopulmonary sound separation because they enable consistent evaluation across different studies. The review includes articles that describe ⁶⁹ datasets of heart, lung, and cardiopulmonary sounds. The studies ⁹⁰ are relevant to the current research because they provide the basis for preprocessing, model training, benchmarking, and evaluation ([Torabi et al., 2025a](#); [Grooby et al., 2022](#); [Ali et al., 2026](#); [Rocha et al., 2023](#)).

¹²⁶ For the current project, it is critical to select the appropriate dataset for the intended application. Ideally, the dataset should include heart and lung sounds or allow the construction of mixed recordings with consistent labels. At the same time, a dataset with restricted or unavailable access is not an option for the current FYP, as the project timeline does not allow for extensive data collection and preparation efforts.

2.6 Evaluation Metrics

A cardiopulmonary sound separation prototype is not something that can be assessed visually, and therefore one has to look beyond the waveform to determine whether the prototype is any good. One might argue that an inspection of the waveform would suffice, but it is recommended that more objective criteria be used for this type of prototype. For example, one may want to check whether the model is able to produce results close to the expected output without interference, and how applicable the prototype is. Processing speeds and the prototype's resource intensity are also important considerations, as the cardiopulmonary separation prototype is meant for student/research purposes.

Finally, it is recommended that performance, source separation, and quality be evaluated to ensure that the prototype actually accomplishes its intended functions. In particular, the review highlights the importance of objective metrics for heart and lung

separation (Torabi et al., 2025a; Roy et al., 2024; Zhao et al., 2024). For the current project, the review serves as the basis for selecting appropriate metrics for the selected dataset and evaluation pipeline.

2.7 Existing Systems and Prototype Relevance

The review includes studies that describe existing algorithms, systems, or software for cardiopulmonary sound analysis, denoising, classification, or separation. The relevance of these articles lies in the detailed description of the approaches that can be used for cardiopulmonary sound separation. However, most of the reviewed studies do not provide end-to-end software that would accept a user upload, validate and preprocess the data, execute the separation model, provide the results, and allow the user to download and review the results. In other words, existing solutions do not provide a prototype in the sense of a single, reusable system.

This finding is critical to the current research since it is a Software Engineering project that requires specification and design, not just an algorithm or a model. In particular, the review highlights the importance of considering requirements, design, interfaces, implementation, and evaluation. Hence, the current review aims to support the identification of requirements for a cardiopulmonary sound separation prototype based on existing literature (Roy et al., 2024; Torabi et al., 2025a; Zhao et al., 2024).

2.8 Literature Review Matrix

Table 1 below provides a brief overview of the studies that were considered for the current research. In particular, the matrix below shows how each study was used in the preparation of the current report. The full literature review matrix is available in Appendix D.

Table 1: Summary literature review matrix

Study	Focus	Method or contribution	Relevance to this project
Grooby et al. (2023)	Noisy neonatal chest sounds	Examines the qualities of noisy recordings and processing requirements	Supports the importance of considering preprocessing and input preparation
Fattahi et al. (2022)	Blind denoising	Denoising strategy for respiratory or	Supports the preparation-focused

		cardiopulmonary recordings	approach and separation prerequisites
Poh et al. (2024)	Neonatal separation	ML-based heart and lung separation	Supports the direction of separation
Sun et al. (2024)	Decomposition-based separation	Separation strategy based on NMF and VMD analysis	Provides a traditional approach baseline direction
Ullah and Zhang (2024)	Unsupervised separation	Unsupervised heart and lung separation approach	Supports the consideration of alternatives
Wang et al. (2023a)	Compact learning model	Compact model for cardiopulmonary processing	Supports the prototyping-focused model
Zhang et al. (2025)	Joint separation approach	DL-based sound separation architecture	Supports the ML-based separation development
Torabi et al. (2025a)	Dataset and benchmarking	Cardiopulmonary sounds dataset / benchmark support	Supports the dataset and evaluation considerations
Roy et al. (2024)	Review study	Review of processing and analysis considerations	Supports the problem framing and research gap analysis
Zhao et al. (2024)	Heart sound analysis	Review or analysis of heart sound processing	Supports the evaluation and relevance analysis

2.9 Research Gaps

Across the reviewed studies, three key gaps were identified. First, cardiopulmonary recordings are notoriously noisy, and mixed heart-lung recordings include artifacts from the recording equipment and the subject's breathing. Second, heart and lung components may overlap, reducing the usefulness of any single-channel recordings. Third, most of the reviewed studies were focused on diagnosis or classification, not on an end-to-end prototype. These gaps are critical to the current research since the cardiopulmonary sound separation prototype will be applied to real-world data with mixed sources and limited training data.

2.10 Summary

The review discusses the studies that formed the basis for the cardiopulmonary sound separation prototype. In particular, the review considers sound characteristics, separation methods, datasets, evaluation metrics, existing systems, and the link

between the previous research and the application-based prototype. The main finding of the review is that the cardiopulmonary sound separation prototype requires noise-aware preprocessing, model-based separation, accessible data, and evaluation planning. The next step in the research is to translate the findings of the literature review into requirements for the cardiopulmonary sound separation prototype.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Overview

The requirements set for the **Machine Learning-Based System for Cardiopulmonary Sound Separation** were identified through a number of steps, including the project scope, literature and document review, current system observation/benchmark, supervisor consultation, and a survey. The last-mentioned fact-finding activity was mandatory because the final solution should be demand-driven and not dictated by the technology. The Google form was distributed, and there were 53 respondents. The responses to section A of the survey are presented below in textual form since this part aims to identify the respondent's profile. On the other hand, sections B to E are depicted in graphical form, and there are 10 graphs to illustrate the requirements because it would be too cumbersome to present 24 questions in textual form. Finally, ⁹⁵the requirements set for the proposed system will be presented.

3.2 Questionnaire

The questionnaire below was designed as a mandatory requirement determination technique to identify the user's needs for the prototype. The items are aimed at obtaining the respondent's practical requirements rather than the theoretical background of such a device. The questions are grouped according to general information about the respondent, their experience in audio or biomedical signal processing, cardiopulmonary audio source separation problems, ideas about the preferred features, interface style, output presentation, and evaluation needs.

The questionnaire consists of 24 items and is provided in **Appendix H**. This section of the paper represents only 10 items, as chapter 3 is not sufficient to cover all details related to this particular requirement determination technique. In other words, not all items from the Google Form are included in this report due to its high volume. However, even though the items are not all presented, they still demonstrate the traceability of requirements.

3.2.1 Questionnaire Design

The questionnaire was designed to identify the requirements for cardiopulmonary sound separation prototype. The target population for the survey comprised of

Software Engineering or Computer Science students, AI or Machine Learning students, Biomedical Engineering or related students, researchers or students with interest in biomedical audio or signal processing, and supervisors or evaluators where applicable.

The questionnaire was posted in Google Forms and 53 responses were received. The introduction informed respondents that the information gathered will be used for research purposes, that there are no right or wrong answers, and that the responses will be kept confidential. The questions were multiple choice, checkbox, Likert scale, and paragraph type.

The structure of the questionnaire is given in **Table 2** below.

Table 2: Questionnaire structure for the user requirements analysis

Section	Purpose	Example question focus	Question type
Section A: Respondent Background	Identify the respondent's background and level of expertise in the field.	Age group, study background, signal processing experience, biomedical audio background, demographic data, and field experience.	Multiple choice
Section B: Current Challenges in Cardiopulmonary Audio Processing	Identify the cardiopulmonary audio processing issues that respondents may encounter.	Separation difficulties, noise, source overlap, recording quality, and absence of simple tools.	Multiple choice, checkboxes, Likert-style scale
Section C: System Feature Preferences	Highlight the prototype features that users may prefer for cardiopulmonary audio processing.	Upload, preprocessing, separation, preview, download, history, and evaluation features.	Multiple choice, Likert-style scale
Section D: Interface and Output Preferences	Identify the interface displays and cardiopulmonary audio result presentations that users may find useful and clear.	Audio player, waveform display, spectrogram display, metrics, interface simplicity, and error messages.	Multiple choice, Likert-style scale
Section E: Evaluation and Improvement	Identify what users expect to evaluate and what	Metrics, processing information, willingness to use	Multiple choice, Likert-style scale,

	improvements they may suggest.	the prototype, and improvement suggestions.	paragraph response
--	--------------------------------	---	--------------------

Table 3 below provides an overview of the questionnaire sections and shows how they were mapped to the 10 items used to structure the analysis in this chapter.

Table 3: Questionnaire analysis mapping

Report item	Main chart used	Related questions used only for discussion	Requirement area supported
Respondent Background and Experience	Section A, Q2-Q5	Section A, Q1 when needed	Respondent profile
Difficulty of Separating Heart and Lung Sounds	Section B, Q1	None	Separation requirement
Problems Affecting Cardiopulmonary Sound Processing	Section B, Q2	Section B, Q3	Preprocessing and separation requirements
Usefulness of Prototype	Section B, Q4	None	System relevance
Preferred System Features	Section C, Q1	Section C, Q6	Functional requirements
Preferred Output After Separation	Section C, Q2	Section D, Q1	Output design
Importance of Audio Preview Function	25 Section C, Q3	Section C, Q4	Preview and download requirements
Preferred Interface Style	25 Section D, Q2	Section D, Q3 and Q5	Usability requirements
First Information Shown After Separation	25 Section Q4	Section C, Q5 and Q6	Result page design
Importance of Evaluation Metrics	25 Section Q1	Section E, Q2	Evaluation requirements
Willingness to Use Prototype	25 Section E, Q3	Section E, Q4	User acceptance and improvements

3.2.2 Respondent Background and Experience

Section A requested for the respondent's background information, for example, age group, respondent category, level of expertise or familiarity with audio/signal processing, biomedical audio experience, and experience with any audio-related tasks. This section is more about describing the respondent's characteristics, and therefore it is summarized here in paragraph form instead of presenting the data in figures.

From the 53 respondents, 39 or 73.6% of the respondents selected the 18-25 years age group, 8 or 15.1% selected the below 18 years age group, and the remaining 6 or 11.3% selected the 26-35 years age group. In terms of the respondent category, 20 or 37.7% respondents identified as Software Engineering or Computer Science students, 15 or 28.3% respondents selected AI or Machine Learning students, 8 or 15.1% selected Biomedical Engineering or related health students, and 7 or 13.2% selected researchers. The remaining respondents selected from other categories including lecturer/supervisor and other respondents.

With respect to the level of expertise or familiarity with audio/signal processing, 29 or 54.7% respondents selected basic, 15 or 28.3% selected not familiar, 6 or 11.3% selected intermediate, and 3 or 5.7% selected advanced. However, 30 or 56.6% of the respondents had prior experience working with biomedical audio, whereas 22 or 41.5% of the respondents did not have any experience. From the audio-related tasks, the most common tasks included audio recording, audio editing, audio classification, audio separation, and machine learning model testing. Overall, the respondent categories are a good mix of users that would benefit from using the proposed system. It is also evident that a significant percentage of respondents would require a simplified prototype interface.

3.2.3 Question 1: Difficulty of ¹¹Separating Heart and Lung Sounds

The first ⁵research question focused on asking the respondents to evaluate the level of difficulty of ⁵separating heart and lung sounds from the mixed cardiopulmonary

recordings. The question aimed to establish whether the cardiopulmonary sounds separation task qualifies as a technical challenge.

1. How difficult do you think it is to separate heart and lung sounds from a mixed cardiopulmonary recording?

53 responses

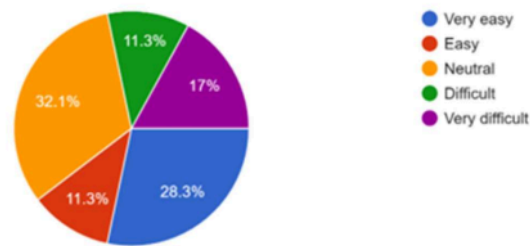


Figure 2: Perceived difficulty of separating heart and lung sounds

Finding: From the 53 respondents that answered this question, the highest number (17 or 32.1%) selected neutral. Furthermore, 15 or 28.3%, 6 or 11.3%, and 6 or 11.3% of the respondents selected very easy, easy, and difficult in that order. Finally, 9 or 17.0% of the respondents selected very difficult. From the finding, it is evident that the cardiopulmonary separation task is not perceived as a challenge at the user level, but rather as a relatively simple task to complete. The finding is an indication that the separation process should be hidden from the user to prevent them from interacting with the underlying technical challenges.

Requirement implication: This finding implies that the system design should hide the cardiopulmonary separation process. Instead, the system should have a streamlined process that allows users to upload a mixed cardiopulmonary audio recording and receive a heart and lung-separated version without having to interact with the separation process.

3.2.4 Question 2: Problems Affecting Cardiopulmonary Sound Processing

The second research question asked the respondents to identify the main problems affecting cardiopulmonary recording or processing. The options included overlapping of heart and lung sounds, challenges in identifying useful sound components, background

noises, low recording quality, poor visual representation of the results, and fewer available software tools.

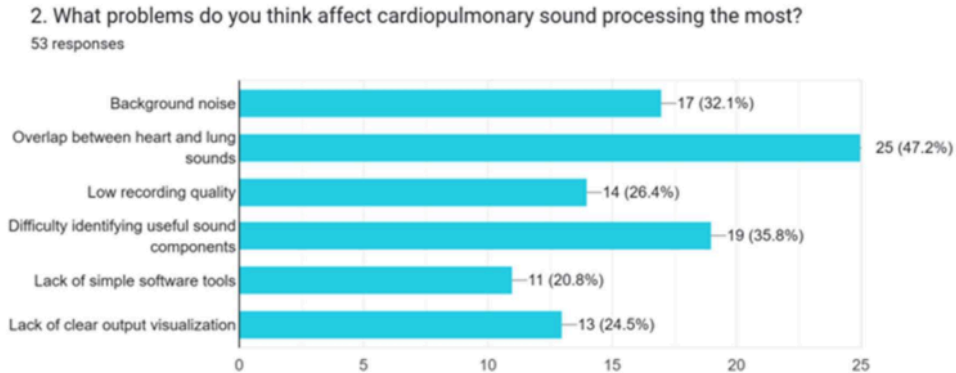


Figure 3: Problems affecting cardiopulmonary sound processing

Finding: This question allowed multiple responses, which is evident from the relatively high percentage of respondents that selected overlapping of heart and lung sounds (25 or 47.2%). Furthermore, many respondents indicated that the inability to easily identify useful sound components (19 or 35.8%), background noises (17 or 32.1%), low recording quality (14 or 26.4%), and the need for better output visualisation (13 or 24.5%) were major problems affecting cardiopulmonary recordings. Similarly, fewer available software tools for cardiopulmonary analysis were also a problem according to 11 or 20.8% of the respondents. In the related noisy audio question, 27 or 50.9% of the respondents either strongly agreed or agreed that noisy audio recordings could affect the accuracy of machine learning-based sound analysis.

Requirement implication: The cardiopulmonary separation prototype needs built-in audio validation, preprocessing, and separation features to address overlapping of heart and lung sounds, background noises, and low-quality audio recordings. The evidence further indicates the need for visualisation features to help the user easily inspect the separated cardiopulmonary recordings.

3.2.5 Question 3: Usefulness of a Heart and Lung Separation Prototype

The third research question asked the respondents whether a simple heart and lung separation prototype would be useful to students, researchers, or users of biomedical audio data.

4. Do you think a simple software prototype for heart-lung sound separation would be useful for students or researchers?

53 responses

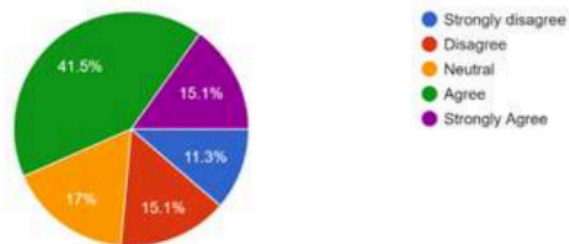


Figure 4: Usefulness of the proposed heart and lung separation prototype

Finding: 22 or 41.5%, and 8 or 15.1% of the respondents selected agree and strongly agree respectively to indicate that a heart and lung separation prototype would be useful. Furthermore, the finding shows that 30 or 56.6% of the respondents selected neutral, disagree, or strongly disagree. However, the overall sentiment is that heart and lung separation prototype would be useful in addressing a cardiopulmonary analysis challenge.

Requirement implication: This finding implies that the cardiopulmonary separation prototype should focus on addressing the cardiopulmonary analysis challenge. As such, it would be useful to focus on an application-oriented design and provide a simple prototype that allows users to upload a mixed cardiopulmonary audio, ³³ separate the audio recording into heart and lung sounds, preview the separated sounds, and download the audio files.

3.2.6 Question 4: Preferred System Features

The fourth research question aimed to identify the system features that would be preferred by the system users. The system features included uploading a mixed cardiopulmonary audio, preprocessing of noisy recordings, separating the

cardiopulmonary audio into heart and lung sounds, preview separated audio, download separated audio, view processing history, and view evaluation results.

1. Which feature would be most important in the proposed system?

53 responses

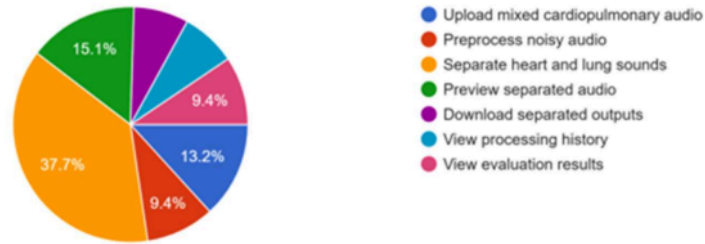


Figure 5: Preferred system features

Finding: The largest percentage of respondents selected separating cardiopulmonary recordings as the most desired system feature (20 or 37.7%). In second and third place were preview separated audio (8 or 15.1%) and uploading a mixed cardiopulmonary audio (7 or 13.2%). The other features received equal percentages of responses (5 or 9.4%) from the respondents. The view processing history feature received 4 or 7.5% of the responses. Similarly, a related history-page question asked the respondents to indicate whether the prototype should allow users to view previous uploads or processing details. The finding indicates a need for a history feature because 44 or 83.0% of the respondents selected yes.

Requirement implication: The cardiopulmonary separation prototype needs to allow users to upload a mixed cardiopulmonary audio, separate the audio into heart and lung sounds, preview the separated audio, and download the separated audio recordings. Furthermore, it would be desirable to allow users to view processing history if the prototype's database can allow tracking of the previously processed cardiopulmonary audio recordings.

3.2.7 Question 5: Preferred Output after Sound Separation

The fifth research question asked the respondents to identify their preferred output after cardiopulmonary separation. The options included receiving only the audio files,

waveform visualisation, spectrogram visualisation, evaluation results, or a combination of any of the available outputs.

2. Which output would you prefer after sound separation?

53 responses

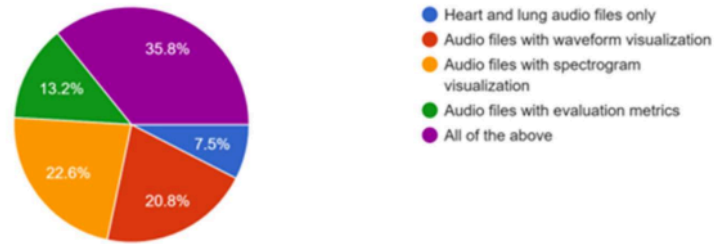


Figure 6: Preferred output after sound separation

Finding: 19 or 35.8% of the respondents selected all the available outputs (audio files, waveform, spectrogram, and evaluation results). Furthermore, 12 or 22.6% and 11 or 20.8% of the respondents selected the audio files with spectrogram and waveform options respectively. In addition, 7 or 13.2% of the respondents selected audio files with evaluation results, whereas 4 or 7.5% selected only the heart and lung audio files. A related result-display question from the cardiopulmonary separation prototype design section showed that 21 or 39.6% of the respondents selected an audio player that displays waveform, spectrogram, and evaluation results.

Requirement implication: The cardiopulmonary separation prototype should not rely on providing only the download link as the output display option. Rather, the prototype should display the separated audio files with options to preview the files as waveforms and spectrograms. In addition, the display should contain the evaluation results.

3.2.8 Question 6: Importance of Audio Preview Function

The sixth research question asked the respondents to evaluate the importance of a preview function for the separated heart and lung sounds before downloading.

3. How important is it for the system to provide a preview of the separated heart and lung sounds?

53 responses

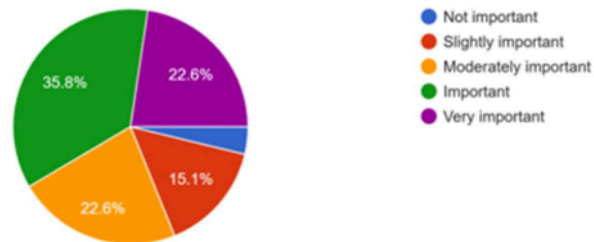


Figure 7: Preview separated heart and lung sounds

Finding: Most respondents (19 or 35.8%) selected the importance of previewing the separated cardiopulmonary audio before downloading. In addition, 12 or 22.6% and 12 or 22.6% of the respondents selected very important and moderately important respectively. Fewer respondents selected slightly important (8 or 15.1%) and not important (2 or 3.8%).

Requirement implication: The cardiopulmonary separation prototype should allow users to preview the separated audio files as a preview function. A download option should also be available for the separated heart and lung audio files.

3.2.9 Question 7: Preferred Interface Style

The seventh research question asked the respondent to indicate their preferred interface style for a cardiopulmonary separation prototype. The options included a simple interface, a detailed interface, and a balanced interface.

2. Which interface style would you prefer?

53 responses

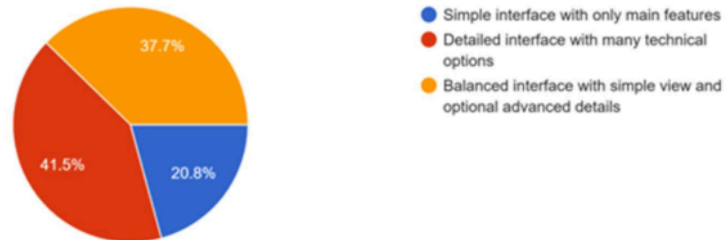


Figure 8: Preferred interface style

Finding: The majority of the respondents (22 or 41.5%) selected a detailed interface with lots of options. A balanced interface with a simple view and optional advanced settings received equal percentages (20 or 37.7%) of responses from the respondents. Finally, 11 or 20.8% of the respondents selected a simple interface with only basic options. A related ease-of-use question in the cardiopulmonary separation prototype design section indicates that the cardiopulmonary separation prototype should also be easy to use as 33 or 62.3% of the respondents selected moderately important, important, or very important options. In the error message question, 29 or 54.7% of the respondents preferred descriptive error messages that explain the cause of an error and suggest ways to address the error.

Requirement implication: The cardiopulmonary separation prototype should focus on providing a good balance between a simple interface (basic options) and a detailed interface (advanced options). As such, the interface can be designed to display the upload and separation process with minimal information on the screen. The interface options such as selection of the audiogram model, preprocessing, evaluation results, waveform and spectrogram should be optional and available on demand. In this way, the prototype would be easy to use while also providing access to additional interface options for more control over the cardiopulmonary separation process.

3.2.10 Question 8: First Information Shown after Separation

117

The eighth research question asked the respondents to indicate the type of information that the cardiopulmonary separation prototype should display after the cardiopulmonary audio has been separated.

4. What information should appear first after separation is completed?

53 responses

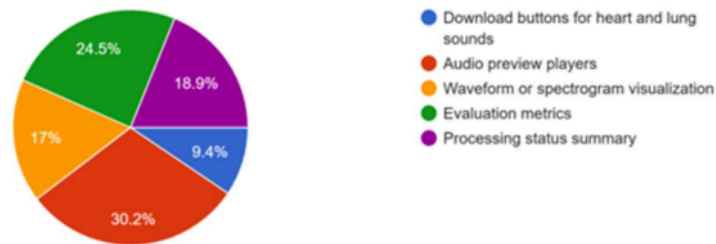


Figure 9: Preferred first information after separation is completed

Finding: Audio preview players were the preferred information to display after cardiopulmonary separation has been completed. This option received 16 or 30.2% of the responses from the respondents. Evaluation results received equal percentages (13 or 24.5%) of responses, whereas processing status summary received 10 or 18.9% of the responses. In addition, waveform or spectrogram received 9 or 17.0%, and download buttons received 5 or 9.4% of the responses. A related processing-information question in the cardiopulmonary separation prototype design section shows that 48 or 90.6% of the respondents rated file name, processing status, and processing time as moderately important, important, or very important.

Requirement implication: The cardiopulmonary separation prototype should display the separated cardiopulmonary audio files with preview players as the first information after the separation process is completed. The second type of information should be the evaluation results followed by the processing status summary, waveform/spectrogram display, and the availability of download buttons for the separated cardiopulmonary audio files.

3.2.11 Question 9: Importance of Evaluation Metrics

The ninth research question asked the respondents to rate the importance of displaying evaluation metrics for the separated cardiopulmonary audio files.

1. How important is it for the system to show evaluation metrics for the separated output?
53 responses

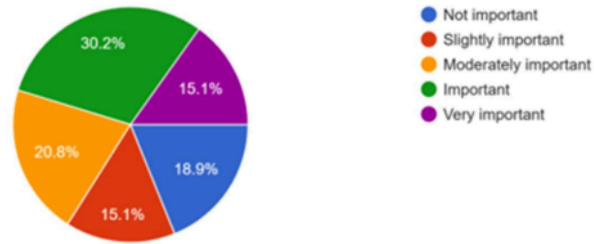


Figure 10: Importance of evaluation metrics for separated output

Finding: 16 or 30.2% and 11 or 20.8% of the respondents selected important and moderately important respectively. In addition, 8 or 15.1% of the respondents selected very important, 8 or 15.1% of the respondents selected slightly important, and 10 or 18.9% of the respondents selected not important. A related evaluation-information question in the cardiopulmonary separation prototype design section showed that the majority of respondents selected all the available information to be displayed when no valid evaluation results are available. Some respondents selected separation accuracy score, processing time, noise reduction information, or signal quality score.

Requirement implication: The cardiopulmonary separation prototype should display evaluation information when valid evaluation results are available. For the FYP1, the evaluation requirements remain a design requirement because it has not been possible to implement a working cardiopulmonary separation model. As such, the cardiopulmonary separation prototype can display processing time, signal quality, separation quality, and noise reduction information to the extent that it is possible given the training data and evaluation dataset.

3.2.12 Question 10: Willingness to Use the Prototype

This question asks whether respondents would be willing to use a prototype that can separate mixed cardiopulmonary audio into heart and lung sound outputs. The related

open-ended question is presented in text below the chart, instead, due to being improvement suggestions rather than specific feedback.

3. Would you use a prototype that separates mixed cardiopulmonary audio into heart and lung sound outputs?

53 responses

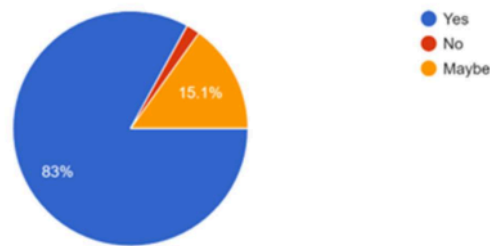


Figure 11: Willingness to use the cardiopulmonary sound separation prototype

Finding: 44 respondents (83.0% of answers) answered yes to the question if they would be willing to use a cardiopulmonary audio prototype able to separate the sounds of the heart and lungs. 8 respondents (15.1%) answered maybe, and 1 respondent (1.9%) answered no. The open-ended question had 18 non-empty responses. The suggestions were varied with some being vague or unrelated to the prototype, but the useful ones included requests for easier use by non-technical users, visualisation of the outputs as waveforms and spectrograms, preview and download features, evaluation metrics, more detailed explanations of results, improved noise reduction features, and better separation quality. Some responses mentioned diagnosis-related features, but they are out of the scope of this FYP, as the prototype only separates sounds without diagnostic capabilities.

Requirement implication: As shown by the high percentage of positive responses among respondents, there is a general willingness to use the described prototype. Therefore, the design should be further explored and developed, but later stages should include improvements suggested by the users. The scope of the prototype should be limited to cardiopulmonary sound separation with additional features such as improved separation quality, usability, visualisation, and evaluation being added in later iterations.

3.3 Functional Requirements

The initial set of the prototype's functional requirements is provided in the table below.

Table 4: Functional requirements for cardiopulmonary sound separation prototype

Req ID	Requirement	Description
F1	Upload Audio File	The system shall accept user's upload of a mixed cardiopulmonary audio file for processing
F2	Validate Audio Format	The system shall perform input file validation, format, size, readability checks before proceeding to processing
F3	Preprocess Audio	The system shall perform any operations necessary to prepare the recording for processing according to the selected workflow, e.g. resampling, normalization, segmentation, noise reduction, etc
F4	Run Cardiopulmonary Sound Separation	The system shall perform the selected cardiopulmonary sound separation pipeline on the input recording using machine learning
F5	Generate Heart Sound Output	The system shall generate the heart output from the input
F6	Generate Lung Sound Output	The system shall generate the lung output from the input
F7	Preview Separated Outputs	The system shall allow the user to preview the separated sound recordings
F8	Download Separated Outputs	The system shall allow the user to download the separated sound recordings
F9	Display Processing Status	The system shall display processing-related information to the user, e.g. file validation results, progress bars, finished or failed processing indicators, etc
F10	Display Evaluation Metrics	The system shall display evaluation-related information if reference signals and/or evaluation metrics are available
F11	View Processing History	The system is required to preserve and display the processing history if storage is available

3.4 Non-Functional Requirements

The initial set of the prototype's non-functional requirements is provided in the table below.

Table 5: Non-functional requirements for cardiopulmonary sound separation prototype

Req ID	Category	Requirement
NF1	Usability	The UI shall be intuitive enough and not require extensive training for the users with different levels of expertise
NF2	Performance	The processing of an acceptable file size shall be performant enough to allow a reasonable demonstration or evaluation of the system's performance
NF3	Reliability	The system shall reliably handle invalid files and failed preprocessing, processing, or output steps and notify the user of the errors
NF4	Maintainability	The system's interface, validation, preprocessing, model, storage, and result management components shall be decoupled to the extent possible to allow future maintenance and updates without affecting the existing functionality
NF5	Security and Privacy	The system shall not store uploaded files on the server if possible and handle the user's data responsibly
NF6	Compatibility	The system shall support a broad range of standard file formats if possible and notify the user of unsupported formats
NF7	Data Integrity	The system design shall prevent integrity-related issues, e.g. overwriting of artifacts by processing the same input file twice
NF8	Scalability	The system shall be designed in a scalable manner to allow the addition of new features, e.g. supporting additional models, visualization options, or evaluation methods, in the future

3.5 User Requirements

The user requirements are summarized in the table below.

Table 6: User requirements for proposed prototype

User Requirement ID	User Need	Description
UR1	Easy audio upload	The user shall be able to easily upload a mixed cardiopulmonary audio file for processing
UR2	File validity feedback	The user shall receive validity feedback about the uploaded file before proceeding to processing

UR3	Separated heart and lung outputs	The user shall be able to obtain separated heart and lung sound outputs after the system has finished processing
UR4	Output preview	The user shall be able to preview the separated heart and lung sound outputs before downloading
UR5	Simple visual feedback	The user shall be able to view simple waveform or spectrogram plots if such displays are supported by the prototype
UR6	Clear status and errors	The user shall receive clear error messages if an error occurs during the file validation and processing steps
UR7	Downloadable results	The user shall be able to download the separated heart and lung sound recordings for review or further analysis
UR8	Basic metrics or processing information	The user shall be able to access basic metric or processing information if supported by the system

3.6 Summary

This chapter used the results of the completed Google Form to define the prototype. Since the form included 53 responses, the results of Section A were summarized while the results of Sections B-E were synthesized into this chapter's 10 requirement-focused tables for easier processing. As a result, it was concluded that the cardiopulmonary sound separation prototype shall process the uploaded mixed cardiopulmonary audio, provide input file validation and preprocessing, separate the input into heart and lung sounds, preview the results, provide downloading options, display processing information, and support evaluation if reference data is available. The requirements were prioritized using the project scope, literature research, supervisor consultation, and requirements questionnaire analysis.

CHAPTER 4: SYSTEM DESIGN

4.1 Overview

The design under consideration is the **Machine Learning-Based System for Cardiopulmonary Sound Separation**, and it is proposed at the prototype level: the ¹¹⁸end-to-end cardiopulmonary sound separation user experience ¹is expected to be simplistic, allowing the user to upload a mixed cardiopulmonary WAV file, choose a separation model (if any), separate the recording, preview the resulting heart and lung sounds, download the separated recordings, and view the separation history.

This design is inspired by one of my Software Design prototype projects in GitHub, which implements the minimal working set of a web interface, backend API, service modules, SQLite metadata storage, filesystem storage for audio data, and model selection. Similarly, the current report proposes analogous concepts for cardiopulmonary sound separation. Most importantly, this design is not expected to demonstrate actual cardiopulmonary sound separation: it defines the prototype to be built for FYP2 but not the separation results to be achieved at FYP2.

4.2 Context Diagram

Figure 12 below represents the context diagram of the proposed cardiopulmonary sound separation prototype. The main external actor is the student/researcher, who uploads the mixed cardiopulmonary ¹input and receives the separated heart and lung sounds in output. The public datasets supply the training data, separation baselines, and evaluation resources for the cardiopulmonary sound separation prototype but are not used by the end user in the minimal prototype version.

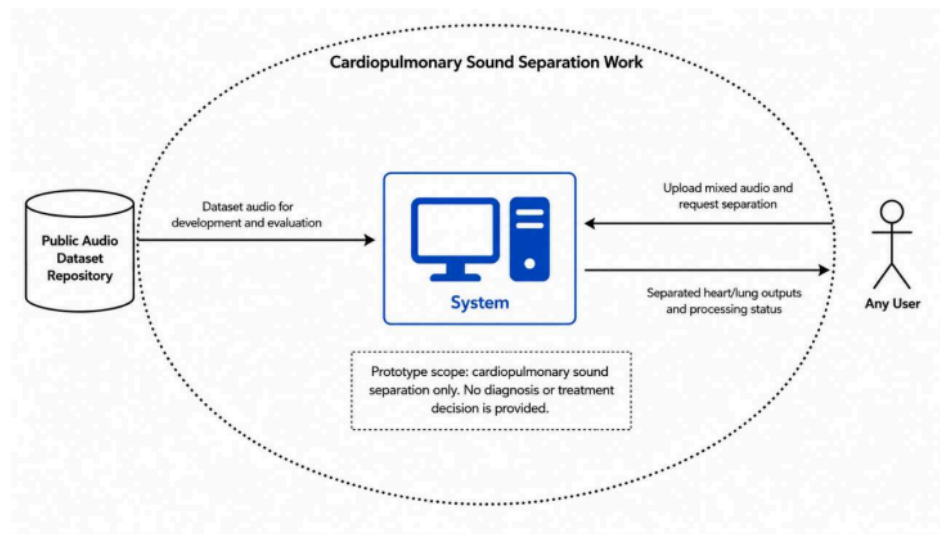
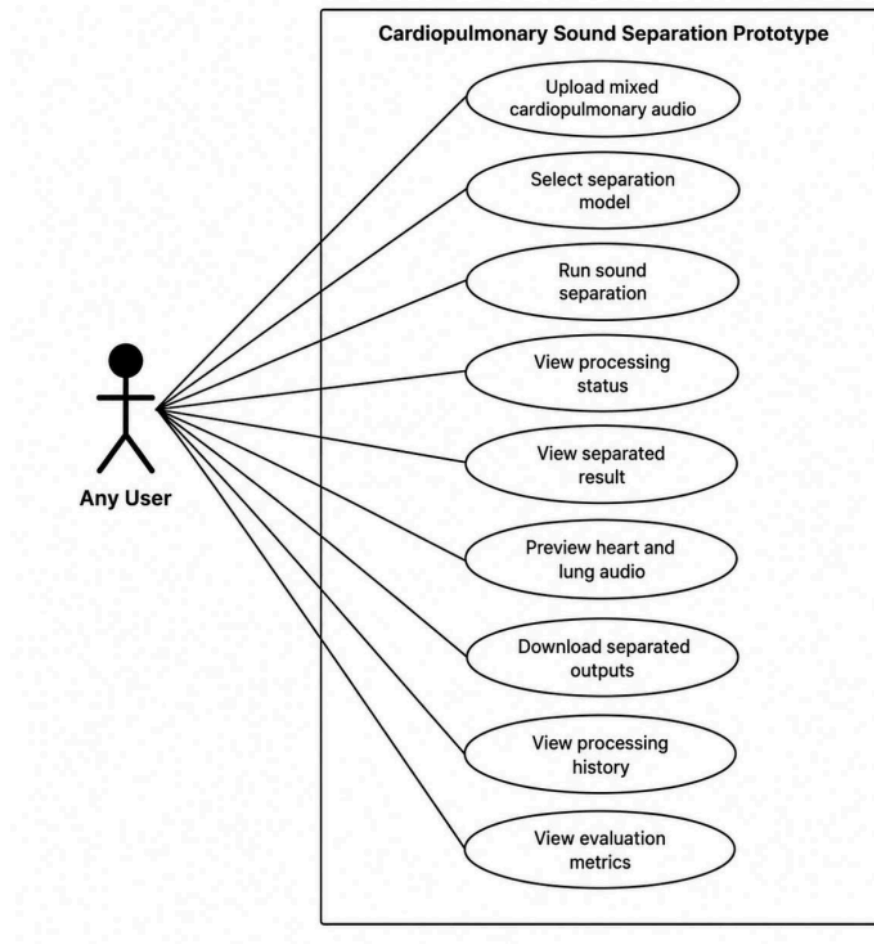


Figure 12: Context diagram for the proposed cardiopulmonary sound separation prototype

The cardiopulmonary sound separation prototype processes the user's mixed cardiopulmonary recording: it validates the input, selects the separation model, performs the separation, and stores/viewing the results. It is not supposed to help out with actual medical diagnosis or treatment, which is why the cardiopulmonary sound separation prototype does not implement disease detection or therapy recommendation features. The design is focused on the mixed cardiopulmonary audio specifics, where centre-based cardiopulmonary sound separation benefits from noise-aware processing of recordings with artefacts, overlapping sources, and diverse signal properties (Grooby et al., 2023; Fattahi et al., 2022; Roy et al., 2024).

4.3 Use Case Diagram

Figure 13 below represents the use case diagram for the cardiopulmonary sound separation prototype. In particular, the proposed design implements the **Select separation model** command rather than upload and separate directly. The reason is that the initial cardiopulmonary sound separation prototype is expected to implement a single separation model, while multiple strategies are foreseen for FYP2.



73 Figure 13: Use case diagram for the cardiopulmonary sound separation prototype

73 4.3.1 Use Case Description

Table 7 below provides a description of the major use case of cardiopulmonary sound separation.

Table 7: Major use case description for the cardiopulmonary sound separation prototype

Item	Description
Use case name	Separate cardiopulmonary sound recording
Primary actor	User (anyone)
Goal	Obtain separated heart and lung sound outputs from one mixed cardiopulmonary recording
Trigger	When the user uploads a supported audio file, selects a model, and initiates the separation procedure
Main output	Heart sound output file, lung sound output file, processing status, and optional metric summary

4.3.1.1 Actors

The cardiopulmonary sound separation use case involves a single external actor: the student or researcher user. They interact with the system to separate mixed cardiopulmonary sounds into heart and lung recordings. The major system components include the web interface, upload manager, model registry, audio validation, preprocessing, separation algorithm(s), result processing, and storage. They are organised in the use case diagram below to define the cardiopulmonary sound separation prototype scope and boundaries but are not actors themselves.

4.3.1.2 Preconditions

The user must possess a cardiopulmonary audio file in a supported format, which the initial prototype version is expected to be WAV. Furthermore, the system must be available to process the request, which includes the database and metadata storage being online and at least one separation model being available. If multiple separation models are available, the interface will offer the user a selection of strategies before proceeding to sound separation. If no strategy is selected, the system will utilise the default cardiopulmonary sound separation model.

4.3.1.3 Normal Flow

1. User opens a prototype interface
2. The system displays a list of available separation models

3. User selects their preferred separation model or leaves the default model active
4. The user selects a separation model or keeps the active default model.
5. User initiates separation request
6. System validates file extension, headers, size, and other metadata
7. System stores the uploaded file and its metadata
8. System pre-processes the audio by converting it into a model-specific waveform
9. System creates a processing job and stores the selected model
10. Separation algorithm factory resolves the chosen algorithm to a strategy
11. Separation engine: processes the sound using the selected strategy, extracting heart and lung sounds
12. Result service stores paths, timing, and status of each processing job
13. Interface displays the result summary, preview controls, and download links
14. User preview/downloads the separated heart and lung sounds

4.3.1.4 Postconditions

As a result of the successful use case execution, the user obtains heart and lung sound recordings. The system stores requested input, model, job status, results data, processing time, and evaluation metrics in case of reference signal presence. In future interactions, if a cardiopulmonary sound separation prototype is updated with a history explorer feature, the user will be able to reconsult the results without uploading the same input recording again.

4.3.1.5 Alternative Flows and Exceptions

In the case of an invalid input file, the system declines the job initiation request before launching the procedure and informs the user of the reason. If the system fails to obtain the list of available separation models, it employs the default strategy if the option is available. If the selected model is not available (lacking checkpoint, configuration, or strategy), the system saves the job with an appropriate model-specific error message. If the preprocessing step fails, the system aborts the job before

initiating the separation algorithm. If the separation algorithm produces less than two signals, the system must not provide the user with an incomplete heart and lung sound set.

4.3.1.6 Non-Functional Requirements

The use case must support usability, reliability, maintainability, performance, privacy, compatibility, and traceability. The interface needs clear status messages because users may not understand backend or machine learning errors. The backend keeps file handling, model selection, strategy execution, result storage, and evaluation as separate responsibilities so FYP2 development can improve one part without rewriting the whole prototype.

4.4 Activity Diagram

Figure 14 represents the planned activity flow. It is noticeable that the available options are loaded before the initiation of the separation process with the user. Therefore, the flow is appropriate for the prototype, where the first working model can be NeoSSNet while later ones are used as comparison alternatives..

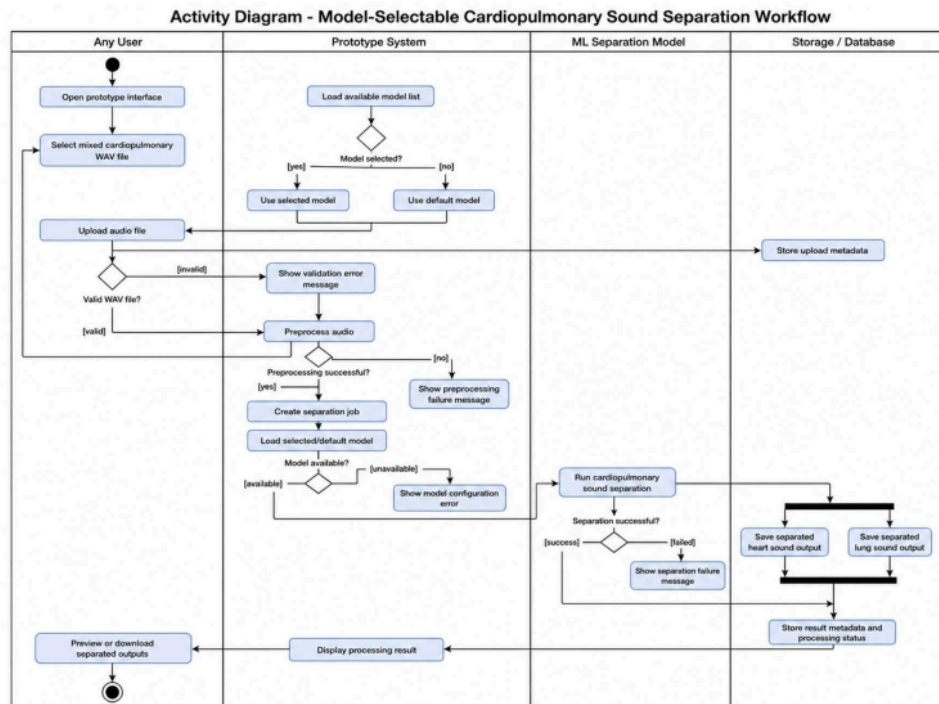


Figure 14: Activity diagram for the sound separation workflow

The activity diagram also specifies the system exit points, where the system is terminated gracefully (e.g., on an incompatible WAV file) or forcefully (e.g., on an unexpected error). Similarly, the system is able to recover from general file upload errors at the beginning of the workflow.

4.5 Class Diagram

Figure 15 below represents the class diagram for the cardiopulmonary sound separation prototype. It follows the service-oriented arrangement, where the interface interacts with route/controller classes, while the processing logic is encapsulated in service classes. In particular, the separation algorithm employs a strategy design pattern to separate mixed heart and lung sounds, which allows future modifications.

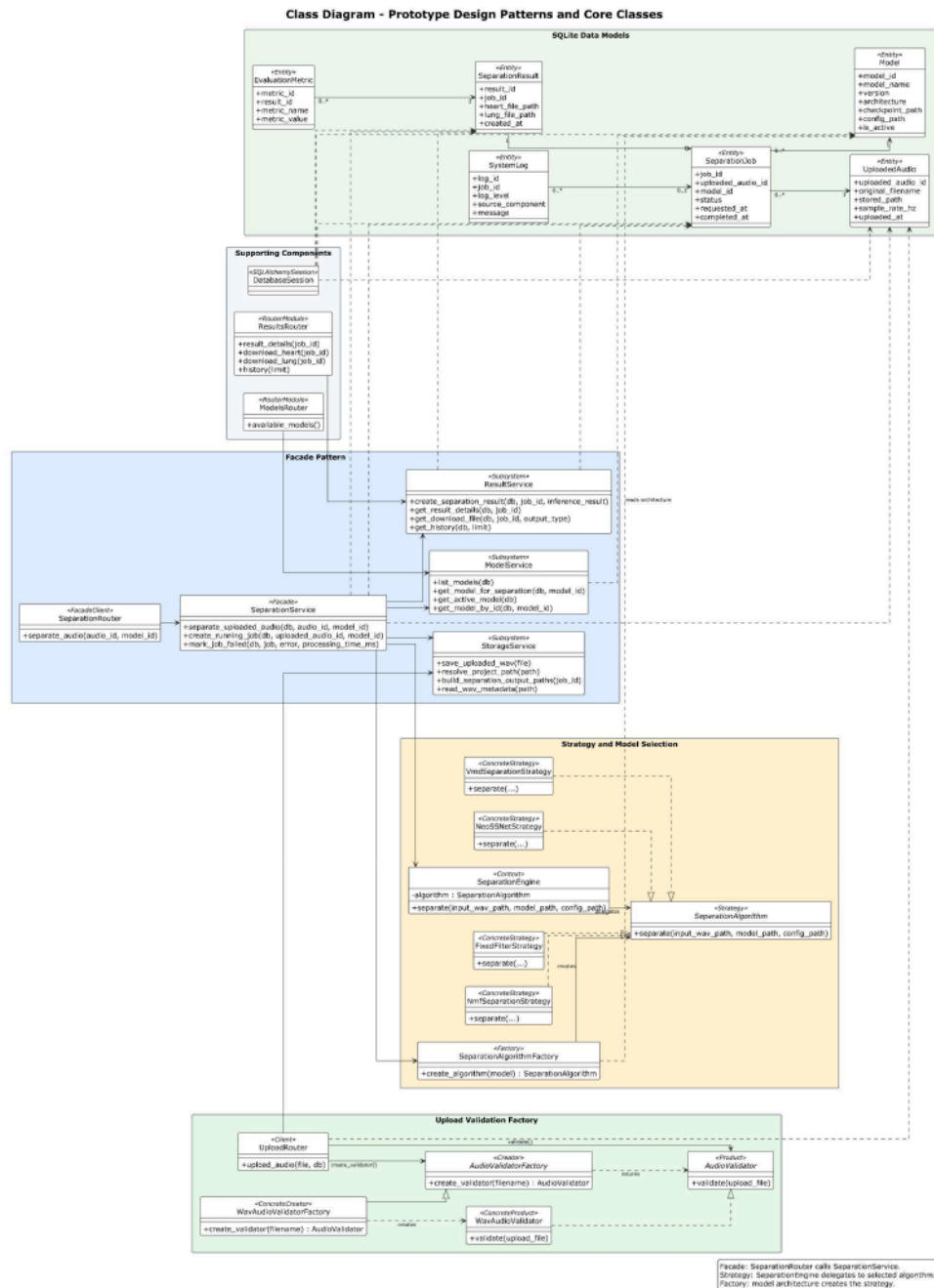


Figure 15: Class diagram for the cardiopulmonary sound separation prototype

4.5.1 Component-Level Architecture

Figure 16 below represents the component-level architecture for the cardiopulmonary sound separation prototype. It incorporates web and API interfaces on the client side, service and model components on the server side, and database and file storage components.

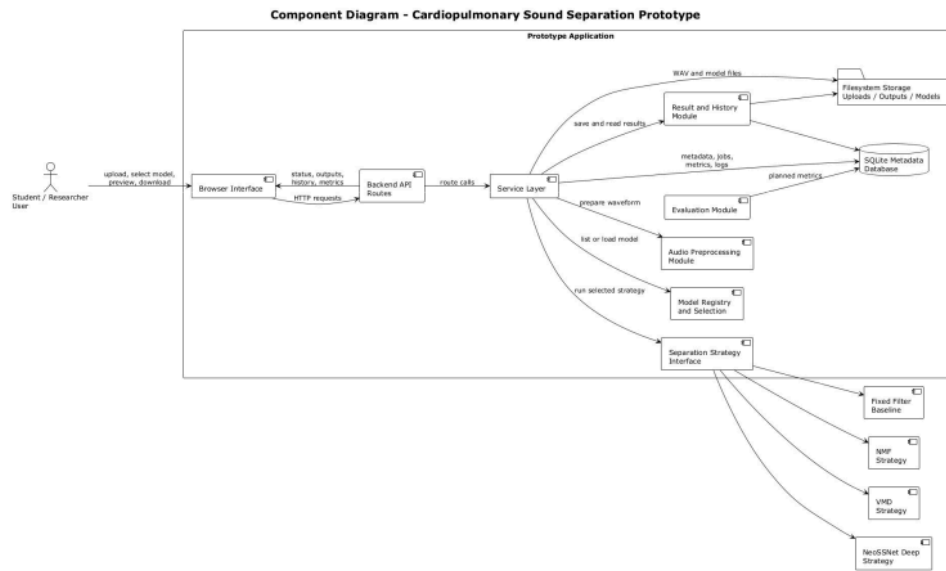


Figure 16: Component-level architecture for the cardiopulmonary sound separation prototype

Table 8 below provides an overview of the cardiopulmonary sound separation prototype components and their responsibilities.

Table 8: Main system modules and responsibilities

Module	Responsibility
User interface	Supplies upload controls, model selection, processing status, result display, preview controls, downloads, and history view
Upload and validation module	Validated WAV input, parsed headers, extracted metadata, and ensured file integrity
Model service	Managed model registry, tracked default model, and validated model checkpoint and configuration paths

Preprocessing module	Converted uploaded audio into a model-specific waveform via mono conversion, resampling, normalisation, segmentation, and time-frequency transformations
Separation service	Coordinated the end-to-end processing workflow from upload to result delivery
Algorithm factory	Abstracted the model-specific implementation details and managed strategy instantiation
Separation strategies	Implemented specific algorithm stratagems, including baseline filtering, NMF/VMD, or NeoSSNet-style deep learning approaches
Result service	Stored result metadata, exposed download locations, retrieved result details, and supplied history information
Database module	Stored model records, job details, result metadata, system logs, and evaluation metrics
Filesystem storage	Saved uploaded mixed audio, generated heart and lung recordings, model files, and system logs

4.5.2 Algorithm and Model Selection Design

Each strategy is expected to follow the same interface to supply flexibility in selecting the algorithm for heart and lung separation. In particular, the strategy is instantiated with a mixed waveform as input and is expected to return two files or two waveforms as outputs: one corresponds to **the heart sound, and the other** corresponds to **the lung sound**. This approach is flexible because it would suffice to implement a single strategy for the initial cardiopulmonary sound separation prototype iteration while leaving the option to expand the model selection in later FYP2 iterations without reworking the interface, result storage, and history display logic.

The algorithm strategies are informed by the literature review on analysis of cardiopulmonary recordings, decomposition methods for mixed signal processing, unsupervised separation methods ¹⁰³ for heart and lung sounds, and the NeoSSNet-style ¹⁷ deep learning paradigm (Grooby et al., 2023; Fattahi et al., 2022; Sun et al., 2024; Ullah and Zhang, 2024; Poh et al., 2024).

Table 9 below lists the strategies to be considered for model selection in FYP2. Each strategy is further described below the table to explain the anticipated contribution to the cardiopulmonary sound separation prototype. Note that the actual results are not described because they will be available in FYP2.

Table 9: Options for model selection in the cardiopulmonary sound separation prototype

Strategy	Role in Prototype	Reason for Inclusion
Fixed filter baseline	Simple baseline to verify that the prototype can process, store, and display separated signals	Limited utility due to overlapping heart and lung responses, but it represents a viable option when these responses are non-overlapping
NMF separation strategy	Fundamental matrix-factorisation baseline or fallback option	Enables comparison with deep learning approaches and provides interpretability
VMD separation strategy	Fundamental mode-decomposition baseline or fallback option	Supplies another decompositional approach for analysing mixed cardiopulmonary recordings
NeoSSNet deep separation strategy	Primary machine learning strategy for heart-lung separation	Aligns with the project title and focuses on the selected direction of research
Future model versions	Optional extension in FYP2 beyond the initial strategy	Allows additional model checkpoints, strategies, or algorithms to be added without complicating the prototype architecture

The model registry will be implemented as a model selection UI, where each registry entry will store the model name, version, architecture, framework, checkpoint path, configuration path, description, and status. Upon model selection, this ⁴⁹information will be stored in the separation job to track the model version. If no model is selected, the system will utilise the default separation strategy.

4.5.3 Database and Storage Design

The database design follows the same principles as the Software Design project: relational tables for metadata, filestorage for binary data. In particular, the cardiopulmonary sound separation prototype will use SQLite as the database management system and the file system for storage, which aligns with the minimal working set prototype approach. This design choice is motivated by the debugging and inspection needs: the cardiopulmonary sound separation FYP1 should be debuggable and inspectable, which is critical for iterative development in FYP1 and FYP2.

Table 10 below provides an overview of the database entities that will be used for the cardiopulmonary sound separation prototype.

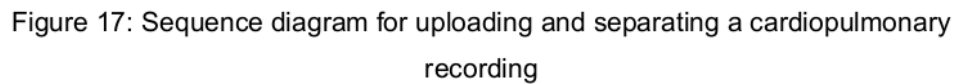
Table 10: Database entities for prototype traceability and analysis

Entity	Description
uploaded_audio	Table to store metadata about each uploaded mixed WAV file, including filename, storage path, sample rate, channels, and duration
model	Table to store information about available models or algorithms, including their names, architectures, checkpoint paths, configuration paths, and statuses
separation_job	Table to track details about each separation request, including the model, status, timestamps, processing duration, parameters, and error messages
separation_result	Table to store metadata about processed heart and lung sound recordings
evaluation_metric	Table to store optional evaluation metrics if reference signals are available for ground truth assessment
system_log	Table to store system-level logs about uploads, processing, results retrieval, and other events

The file system will be organised into subdirectories to store uploaded files, temporary files, heart sound recordings, lung sound recordings, model files, and system logs. This organisation will supply flexibility for data inspection and cleanup procedures because it will be relatively straightforward to delete demonstration data and differentiate public test data from prototype files.

4.6 Sequence Diagram

Figure 17 below represents the sequence diagram for cardiopulmonary sound separation. It specifies the interactions between the interface and the underlying modules. In particular, the model registry needs to obtain the list of strategies before the user attempts to upload an audio file. As the user selects a strategy, the separation strategy is instantiated by the separation algorithm factory. Finally, upon separation completion, the result service supplies the result details to the interface.



4.7 Interface Design

Table 11 below summarises the interface areas for cardiopulmonary sound separation.

Table 11: Interface areas for cardiopulmonary sound separation

Interface Area	Description
Upload area	WAV file selector, note about the accepted format, selected filename, and Start button
Model selection area	Dropdown list with available separation strategies, default model indicator, and strategy description
Status area	API status, upload validation status, preprocessing status, model execution status, completion time, and error messages
Result area	Preview controls for heart and lung sounds, result file download controls, and optional evaluation display
History area	List of recent jobs with filename, status, duration, and options to reopen or download results

No interface mockups are produced for this FYP1 report because interface design and user testing belong to FYP2. However, the interface areas inform the questionnaire in Chapter 3, particularly the result preview options, file download instructions, status displays, and evaluation viewport.

4.8 Summary

The cardiopulmonary sound separation design under consideration introduces the prototype workflow, where the user uploads a mixed recording, selects a separation model, and obtains separated heart and lung recordings as result files. The design also specifies the separation model registry, algorithm strategies, database, and storage design to isolate these elements for future iterations. The next chapter builds upon this prototype description, outlining the cardiopulmonary sound separation implementation and testing to inform the FYP2 prototype without overrating the separation results at the FYP1 level.

CHAPTER 5: IMPLEMENTATION PLAN

5.1 Development Phase

The development phase is expected to focus on implementing the requirements and design specified in FYP1 as a prototype in FYP2. The proposed prototype is a web application with a browser-based interface, API, service layer, light database, file storage, and model selection. This chapter will specify the implementation intentions. In particular, it does not include completed separation accuracy, user testing, or implementation quality control. Regarding implementation principles, the separation algorithms are expected to be added on top of the common application workflow. Upload, validation, model selection, job status, storage, preview, download, and history are general functions that can be reused across different separation algorithms. Particular algorithms, in their turn, can be switched via the common strategy interface.

5.1.1 Front-End Development

The front end is supposed to provide a user interface for cardiopulmonary audio upload, model selection, validation, process status, result preview, and file download. Its development is considered reasonably straightforward, as the target audience is students and researchers, who are used to working with such tools. In particular, there is no need to complicate the interface with medical-specific terminology or settings, as the tool is not intended to be used in a clinical setup.

Instead, the following areas are going to be implemented:

1. The upload area: a file selector for WAV file upload and process initiation
2. The model selector: an element allowing the user to switch between separation algorithms or default models
3. The process status area: information about the upload progress, validation status, model initiation, process progress, and possible errors
4. The result preview area: a preview of separated heart and lung sounds and respective download buttons
5. The job history area: a list of completed processes with the possibility to re-open them

The front end is supposed to be intuitive, making sure that the user knows that the tool only separates sounds and is not intended to provide medical-level analysis. Also, the job status area is supposed to clearly signalize about the most common errors, such as unacceptable file format, absence of a model, preprocess failure, separation failure, and others.

88

5.1.2 Back-End Development

The back end shall process the uploaded files, validate the audio, choose and execute the separation algorithm, store and retrieve the results, and inform the front end about the status of the process. The back end is planned to be a thin controller with route-specific services (see **Table 12**). The routing logic is responsible for processing the request and returning an HTTP response, and the services define the business logic of file storage, model selection, job lifecycle, and result retrieval.

The table below outlines the planned API endpoints and their purpose.

Table 12: Planned backend API responsibilities

Endpoint or function	Purpose
GET /health	Check if API and database are online
GET /models	Get list of available models and default configuration
POST /upload	Upload file and validation
POST /separate/{audio_id}	Execute separation
GET /result/{job_id}	Get job status, metadata, links to results and timing
GET /download/{job_id}/heart	Download separated heart
GET /download/{job_id}/lung	Download separated lung
GET /history	Get history of recent jobs for history screen in UI

The back end is supposed to decouple model execution and routing, so that FYP2 would be able to implement the alternative strategy and update the model execution independently from the upload and result APIs.

5.1.3 Machine Learning Model Integration

The machine learning model will be integrated using the strategy pattern, which will be implemented at the back end. Each strategy will receive the mixed input waveform and return the ¹¹separated heart and lung sounds according to the same contract. This will allow the tool to implement the baseline algorithm, conventional signal separation approaches, and the machine learning model as interchangeable strategies. The separation strategies will be based on the reviewed articles (³⁴Fattahi et al., 2022; Sun et al., 2024; Ullah and Zhang, 2024; Poh et al., 2024).

In particular, ⁷⁶the mixed cardiopulmonary signal $x[n]$ can be expressed as:

$$x[n] = h[n] + l[n] + r[n],$$

⁶⁰where $h[n]$, $l[n]$, and $r[n]$ are the heart, lung, and residual components, respectively. The goal of the separation strategy is to estimate $h[n]$ and $l[n]$ given the mixed $x[n]$.

5.1.3.1 Input Preparation

The input preparation step is essential to maintain uniformity: the recorded cardiopulmonary sounds can contain a variety of noise, artefacts, and preparation issues, which will undermine the separation process (Grooby et al., 2023; Fattahi et al., 2022).

A generic input preparation workflow could look like this:

1. Read the uploaded file.
2. Convert stereo files to mono if necessary.
3. Resample to the target sample rate.
4. Perform amplitude normalization.
5. Apply segmentation or transformation if necessary.

Amplitude normalization could be performed according to ⁹⁷the following formula:

$$\tilde{x}[n] = \frac{x[n]}{\max_n |x[n]| + \epsilon},$$

where $\text{amp_scale} = \text{max_amp} / (\text{max_amp} - \text{min_amp} + \text{EPS})$ with EPS being a small value to avoid division by zero. Short-time Fourier transform (STFT) can be applied if the separation strategy requires a time-frequency representation:

$$X(k, m) = \sum_{n=0}^{N-1} \tilde{x}[n]w[n - mH]e^{-j2\pi kn/N},$$

where $w[n]$ is the window function, H is the hop size, N is the FFT size, k is the frequency bin, and m is the frame index.

5.1.3.2 Algorithm 1: Fixed Filter Baseline

The fixed filter baseline is a simple approach that can be used as a comparison baseline and a preparatory test for more involved decompositions (Roy et al., 2024; Zhao et al., 2024). It implements rough frequency filtering with fixed masks or filters to isolate the heart and lung sounds:

$$\hat{H}(k, m) = M_h(k)X(k, m),$$

$$\hat{L}(k, m) = M_l(k)X(k, m),$$

with $(M_h(k))$ and $(M_l(k))$ being the fixed masks or filters for the heart-dominated and lung-dominated frequencies, respectively. The time-domain estimates are reconstructed using the inverse STFT:

$$\hat{h}[n] = \text{iSTFT}(\hat{H}(k, m)), \quad \hat{l}[n] = \text{iSTFT}(\hat{L}(k, m)).$$

This algorithm is not expected to perform well on strongly overlapping cases, but the actual utility of such a trivial algorithm is to sanity check that the upload, preprocessing, output writing, preview, and download procedures work before proceeding to a more involved model.

5.1.3.3 Algorithm 2: NMF and VMD-Based Decomposition

Non-negative matrix factorization (NMF) and variational mode decomposition (VMD) can be suggested as two particular traditional approaches. The decomposition methods are suitable for providing explainable baselines for cardiac-lung separation and comparison (Sun et al., 2024; Ullah and Zhang, 2024). They could be implemented

in the proposed design as NmfSeparationStrategy and VmdSeparationStrategy classes or particular cases of the same interface.

For the NMF strategy, if $(V = |X|)$ is the non-negative magnitude spectrogram, NMF approximates it as:

$$V \approx WH,$$

where (W) contains spectral basis patterns and (H) contains their time activations. A common reconstruction loss takes the form of:

$$\min_{W, H \geq 0} D(V || WH) + \lambda R(W, H),$$

where $D(\cdot)$ is a divergence measure and $R(\cdot)$ is a regularisation term. Separate heart- and lung-specific estimates can be obtained by defining soft masks:

$$M_h = \frac{V_h}{V_h + V_l + \epsilon}, \quad M_l = \frac{V_l}{V_h + V_l + \epsilon}.$$

and separate the contents of the spectrogram by:

$$\hat{H} = M_h \odot X, \quad \hat{L} = M_l \odot X,$$

Decomposition can then be applied to the time series directly to isolate the heart and lung sounds. Alternatively, VMD can be applied to the signal $x[n]$ to obtain the modes $u_k(t)$ with centre frequencies f_k :

$$\min_{\{u_k\}, \{\omega_k\}} \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2$$

such that:

$$\sum_k u_k(t) = x(t).$$

The modes can then be separated into the heart- and lung-specific components by grouping the ones with similar centre frequencies.

5.1.3.4 Algorithm 3: NeoSSNet Deep Separation Strategy

NeoSSNet-style deep separation strategy will be the core machine learning model, as it was designed for the cardiopulmonary separation and can be employed for this task directly ([Poh et al., 2024](#)). It is expected to utilise the input from the preparatory steps and estimate the two output channels with a neural network:

$$[\hat{h}, \hat{l}] = f_{\theta}(\tilde{x}),$$

where f_{θ} represents the model with parameters θ . If the separation strategy involves time-frequency masks, they can be represented as:

$$[M_h, M_l] = g_{\theta}(X),$$

$$\hat{H} = M_h \odot X, \quad \hat{L} = M_l \odot X.$$

If the references are available for the training data or the evaluation set, a reconstruction loss can be defined as:

$$\mathcal{L}_{rec} = \|h - \hat{h}\|_1 + \|l - \hat{l}\|_1.$$

For cardiopulmonary separation, the SI-SDR between the estimate y and the reference source s can also be used if the relevant information is available ([Torabi et al., 2025a](#); [Zhao et al., 2024](#)). For one estimated source ($\hat{s}$) and reference source (s):

$$s_{target} = \frac{\langle \hat{s}, s \rangle}{\|s\|^2} s,$$

$$e_{noise} = \hat{s} - s_{target},$$

$$SI - SDR = 10 \log_{10} \frac{\|s_{target}\|^2}{\|e_{noise}\|^2}.$$

The first FYP2 target is working inference path from upload to separated output files. After that, the training, fine-tuning, or model replacement can be attempted if enough data, suitable hardware, and supervisor's approval are available.

5.1.3.5 Model Selection Strategy

A model choice strategy can be implemented as a strategy to select the algorithm to use on upload time: the user will be able to choose the model from a list. The controller

will pick the model-specific strategy if present in the request or use the default one. Factory methods will be used to tie the model configuration to the strategy:

```
strategy = Factory(model.architecture).
```

For example, for a specific architecture NeoSSNet, you can make a model record, which would create a NeoSSNet strategy, and specific names FixedFilter, NMF, VMD would create their own concrete strategies. By this approach, you can easily extend the system by adding new algorithms and strategies - it will require only creating a new strategy class and entry in the model registry, instead of changing the whole application.

5.1.4 Database and Storage Integration

The prototype will utilize light-weight database storage for metadata and file system storage for audio files. The database should store upload metadata, model metadata, selected model ID, job status, output paths, processing time, optional evaluation metrics, and system logs. On the file system, there should be storage for audio files, model files, and outputs.

The storage folders are:

```
storage/uploads/raw/  
storage/uploads/temp/  
storage/outputs/heart/  
storage/outputs/lung/  
storage/ml_models/  
storage/logs/
```

This design is practical and a good choice for a local academic prototype because it minimizes the database's size footprint while still allowing traceability of uploaded files, selected model records, processing jobs, and resulting outputs.

5.2 Testing

The test phase will focus on ensuring that the prototype works as designed and that the individual strategies are safe to use within the application. The test activities are

described below, but they are planned rather than executed: the actual test execution will happen in FYP2.

25

5.2.1 Unit Testing

The **unit tests** will be **written** for generic modules and individual strategies. In particular, WAV validation, metadata extraction, path management, model lookup, default model selection, and strategy factory will be added. The separation strategy unit tests should not be checking for specific separation quality. Instead, they should make sure that the strategy correctly initiates and terminates in case of an incorrectly prepared input. The tests should also ensure that the strategy returns correct file paths for heart and lungs sounds and does not falsely report a finished job.

5.2.2 Integration Testing

The integration tests will ensure that the front end, API, database, storage, preprocessing, model selection, strategy factory, separation process, and result retrieval are compatible and work together. The following cases will be covered:

1. A user uploads a WAV file, and the metadata is stored in the database.
2. The model list is requested, and the default model is selected.
3. A separation process is initiated with a specific model ID.
4. The job records the selected model.
5. The successful job has heart and lung output paths recorded in the database.
6. The result and download endpoints return the correct job information.
7. The failed job with missing model files or an unsupported model architecture returns an informative error message.

5.2.3 System Testing

The system tests will follow the user interaction scenarios and attempt to repeat them to ensure that the prototype behaves as expected. In particular, the tests will include valid upload, unsupported file upload, invalid WAV header, model list retrieval, model-specific job execution, default model fallback, result preview, output download, history retrieval, and failure recovery. The tests will also check the user interface elements: the

progress notifications, failed job display, unavailable download buttons, and the absence of medical diagnosis language.

5.2.4 User Acceptance Testing

User acceptance tests can be conducted once the prototype is completed, and actual users are available to participate. The test scenarios will involve uploading a record, selecting a model, reviewing the result and checking the contents of the result page. This will verify if the model selection process is intuitive, the file is not downloaded before the research process is completed, and the result page provides sufficient information for research. No user acceptance tests have been carried out yet, as the prototype is not ready at this stage of development.

5.3 Deployment

The deployment phase will prepare the prototype for presentation and evaluation. The initial deployment target is the local machine because it will entail fewer security and hosting considerations: the tool will only process the uploaded cardiopulmonary recordings, separate them with the selected model, and store the results on the file system. The deployment instructions will describe how to set up the environment, launch the prototype, process an example WAV file, and review the results. In particular, the following steps are planned:

1. **Setting up the Python environment.**
2. **Installing the backend and all the necessary machine learning libraries.**
3. Preparing a database.
4. Making sure the storage location is available.
5. Putting the weights and configuration files in the working directory.
6. Starting the backend and the browser interface.
7. Uploading a short WAV file as a test.
8. Checking the progress and making sure that heart and lung files have been created.

If the prototype is later deployed elsewhere, additional work will have to be performed to host the API, secure the file storage, limit access to sensitive parts, manage long-running jobs, and monitor performance. These topics are outside the scope of the initial FYP1 effort, but they can be noted as future considerations.

5.4 Documentation and Maintenance Plan

I will document the deployment process, provide notes for end-users, describe how the API should be used, cover test coverage notes in the documentation, and maintain a distinction in the report between what was implemented, what was intended but not implemented, and what remains planned. The maintenance will involve updating the libraries and dependencies, providing more error context, implementing separation strategies, improving model selection, updating the model weights, improving preprocessing, implementing waveform display, evaluating the separation, and performing other toolbox-specific maintenance tasks.

5.5 Summary

The development plan for the FYP2 implementation was designed around the practical limitation of time, aiming to achieve a working prototype that would not exaggerate its accomplishments. In this chapter, I described the front-end, API, model selection and algorithm strategies, database and file storage, testing, deployment, documentation, and maintenance plans as the tasks for FYP2. The cardiopulmonary separation strategies I intend to implement in FYP2 include the baseline fixed filter, decomposition using NMF and VMD, and the separation using the NeoSSNet-style machine learning model. This chapter focused on the FYP1 task of planning rather than implementing cardiopulmonary separation strategies. Therefore, it does not describe the testing outcomes, signal separation quality, model evaluation, or user experience tasks that have not been completed yet.

99 CHAPTER 6: CONCLUSION

6.1 Overview

This is the concluding chapter for the cardiopulmonary sound separation project based on machine learning study, as part of FYP1 report. The background study, literature review, requirements analysis, system design, and implementation plan for FYP2 are presented and discussed.

6.2 Project Summary

The cardiopulmonary sound separation project aims to solve a particular problem of mixed heart and lungs sounds recordings, from which the separate heart and lungs sounds need to be extracted. The literature review showed that the recordings may include various artefacts, noises, and overlapping components of interest. Thus, using a traditional signal processing approach for filtering and extraction may not be efficient enough; more involved machine learning algorithms may be required to identify and separate the patterns, given the appropriate database and evaluation criteria.

The cardiopulmonary sound separation prototype to be developed during FYP1 will be based on the workflow that includes upload, validation, preprocessing, selection of model, separation, preview, and download or export options. The requirements analysis and system design have been performed so that the prototype scope can be better defined and developed during FYP1.

6.3 Expected Contributions

The cardiopulmonary sounds separation prototype is expected to contribute to the body of knowledge by providing a proof-of-concept software tool for the separation of the heart and lungs sounds from the recordings of cardiopulmonary sounds. The tool is aimed at the student/researcher who needs to process the mixed recordings and extract the separate heart and lungs sounds for further analysis.

Regarding the expected contribution within the scope of FYP1, my primary contribution is a background study, requirements analysis, design artefacts, and implementation plan that were produced to support the FYP2 research and development.

6.4 Limitations

FYP1 findings represent only the beginning stage of the research, which means that my contribution to the final prototype is minimal and is limited to the requirements analysis. Moreover, the findings described in this report are preliminary in nature and do not contemplate the implementation, testing, performance, and user evaluation stages of the prototype. Additionally, the proof-of-concept tool's performance will depend on the availability of public databases, quality of recordings/stimuli, model performance, and time available for FYP2 implementation.

I wanted to make it clear that the intended contribution is not to provide a clinical diagnosis/treatment recommendation. Cardiopulmonary-sounds separation prototype should rather be seen as a research tool to evaluate the feasibility of this approach for the specific database within the domain of clinical medicine.

6.5 Future Work

The future work within FYP2 consists of building the cardiopulmonary-sounds separation prototype according to the design specifications, preparing the public database/recording set, performing the separation, preview and download/export steps, and evaluating the performance outcomes. Regarding the potential contribution of future work, the findings of the questionnaire provided in Chapter 3 can be used to scope the prototype preview and download/export options, statuses, messages, and other features identified by end-users during the requirement analysis. Other ideas for future research include developing the prototype's user interface, supporting multiple recording file formats, improving the preprocessing, model selection, performance visualisation, and providing a more comprehensive comparison with the baseline approaches.

6.6 Summary

The cardiopulmonary-sounds separation project for the machine learning module is summarised in this concluding chapter, with a focus on FYP1 report. The cardiopulmonary-sounds-separation prototype design and implementation steps were demonstrated. Following the FYP1 prototype planning, the cardiopulmonary-sounds-separation project will proceed to implementation, testing, and evaluation stages that will take most of the remaining time during FYP2.

References

1. S. N. Ali, S. B. Shuvo, M. I. S. Al-Manzo, A. Hasan, and T. Hasan. (2023). An end-to-end deep learning framework for real-time denoising of heart sounds for cardiac disease detection in unseen noise. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2023.3292551>.
23. S. N. Ali, A. Zahin, S. B. Shuvo, N. B. Nizam, S. I. S. K. Nuhash, S. S. Razin, 6. M. S. Sani, F. Rahman, N. B. Nizam, F. B. Azam, R. Hossen, and S. Ohab. (2026). BUET multi-disease heart sound dataset: A comprehensive auscultation dataset for developing computer-122 aided diagnostic systems. *Computer Methods and Programs in Biomedicine Update*. <https://doi.org/10.1016/j.cmpbup.2026.100237>.
15. Y. Arjoun, T. N. Nguyen, T. Salvador, A. Telluri, J. C. Schroeder, R. L. Geggel, J. W. May, D. K. Pillai, S. J. Teach, S. J. Patel, R. W. Doroshov, and R. Shekhar. (2023). StethAid: A digital auscultation platform for pediatrics. *Sensors*. <https://doi.org/10.3390/s23125750>.
1. R. Baeyens, D. Ragusa, T. Stas, K. Ides, E. Marenzi, F. Leporati, W. Daems, and J. Steckel. (2025). Compressed sensing of acoustic cardiopulmonary signals using a CNN-based reconstruction method. *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. <https://doi.org/10.1109/embc58623.2025.11253482>.
36. S. Behera, P. K. Jain, and S. Bakshi. (2026). Adaptive thresholding of DWT coefficients using UNet for 51 denoising real-life respiratory sounds. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2025.108232>.
3. D. Fattahi, R. Sameni, E. Grooby, K. Tan, L. Zhou, A. King, A. Ramanathan, A. K. Malhotra, and F. Marzbanrad. 3. (2022). A blind filtering framework for noisy neonatal chest sounds. *IEEE Access*. <https://doi.org/10.1109/access.2022.3170052>.
- E. Grooby, C. Sitaula, D. Fattahi, R. Sameni, K. Tan, L. Zhou, A. King, A. Ramanathan, A. Malhotra, G. A. Dumont, and F. Marzbanrad. (2022). Real-time multi-level neonatal heart and lung sound quality assessment for telehealth applications. *IEEE Access*. <https://doi.org/10.1109/access.2022.3144355>.
5. E. Grooby, C. Sitaula, D. Fattahi, R. Sameni, K. Tan, L. Zhou, A. King, A. Ramanathan, A. Malhotra, G. Dumont, and F. Marzbanrad. (2023). Noisy neonatal chest sound separation for high-14 quality heart and lung sounds. *IEEE Journal of Biomedical and Health Informatics*. <https://doi.org/10.1109/JBHI.2022.3215995>.
41. B. Han and W. Quan. (2025). Cardiorespiratory sound separation using singular spectrum 121 analysis. *2025 17th International Conference on Signal Processing Systems (ICSPS)*. <https://doi.org/10.1109/icsps66615.2025.11347745>.
28. V. T. Hoang, Q. Nguyen, Q. C. Nguyen, X. P. Nguyen, and H. Nguyen. (2023). Sound-dr: Reliable sound dataset and baseline artificial intelligence system for respiratory illnesses. *PHM Society Asia-Pacific Conference*. <https://doi.org/10.36001/phmap.2023.v4i1.3604>.

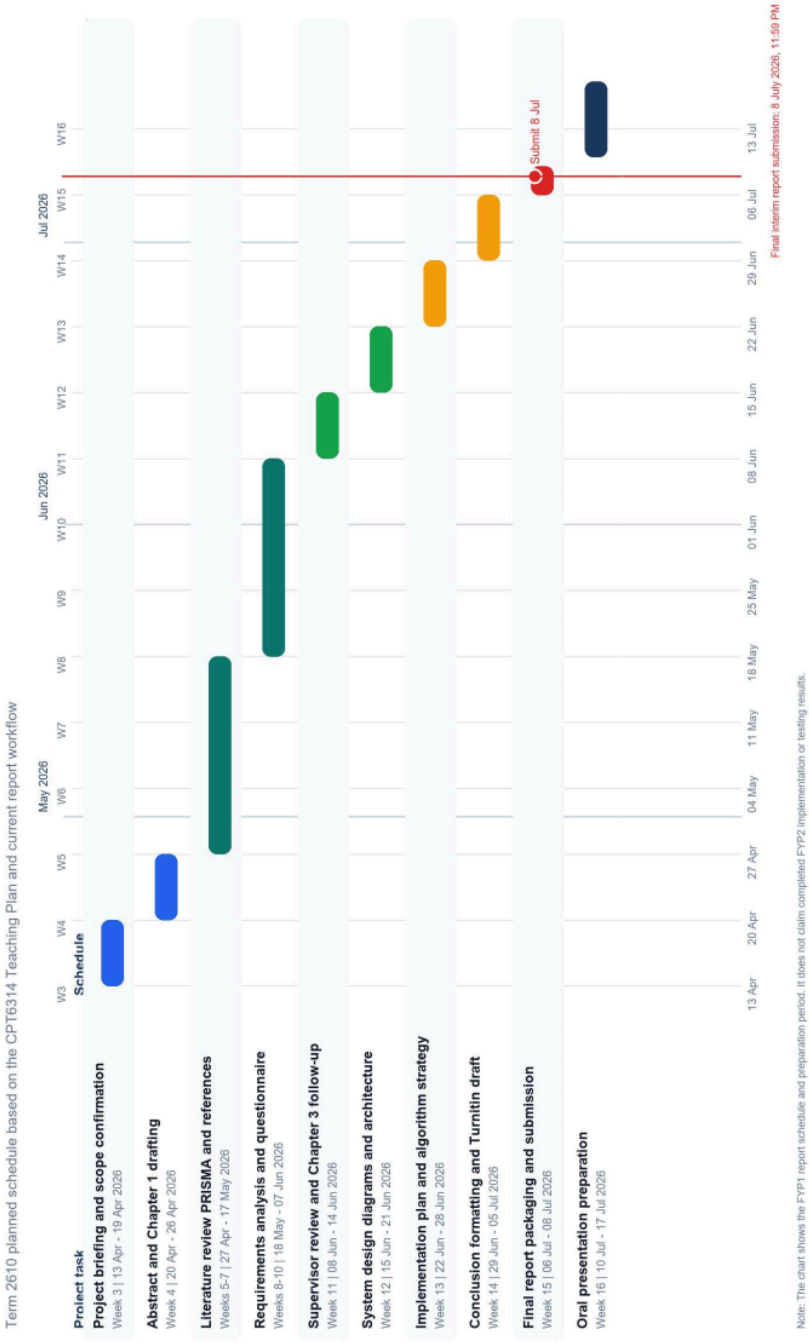
- M. Jakubec, E. Lieskovska, and P. Pocta. (2025). A robust deep learning based model for denoising phonocardiogram signals in clinical environments. *Engineering Applications of Artificial Intelligence*. <https://doi.org/10.1016/j.engappai.2025.112286>.
- E. Mutlu, V. Hüseyin, and G. Serbes. (2024). Bridging auscultation and tiny machine learning: A digital stethoscope leveraging convolutional neural networks on an embedded device for organ sound analysis. *Traitement du Signal*. <https://doi.org/10.18280/ts.410521>.
- R. Pal and A. Barney. (2024). Automatic breathing phase identification based on the second derivative of the recorded lung sounds. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2024.106315>.
- Y. Y. Poh, E. Grooby, K. Tan, A. Malhotra, M. Harandi, and F. Marzbanrad. (2026). Reprogramming automatic speech recognition models for neonatal chest sound separation. *IEEE journal of biomedical and health informatics*. <https://doi.org/10.1109/jbhi.2026.3669531>.
- Y. Y. Poh, E. Grooby, K. Tan, L. Zhou, A. King, A. Ramanathan, A. Malhotra, M. Harandi, and F. Marzbanrad. (2024). NeoSSNet: Real-time neonatal chest sound separation using deep learning. *IEEE Open Journal of Engineering in Medicine and Biology*. <https://doi.org/10.1109/OJEMB.2024.3401571>.
- M. F. Pouyani, M. Vali, and M. A. Ghasemi. (2022). Lung sound signal denoising using discrete wavelet transform and artificial neural network. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2021.103329>.
- M. A. Reyna, Y. Kiarashi, A. Elola, J. Oliveira, F. Renna, A. Gu, E. A. P. Alday, N. Sadr, A. Sharma, J. Kpodonu, S. Mattos, M. T. Coimbra, R. Sameni, A. B. Rad, and G. D. Clifford. (2023). Heart murmur detection from phonocardiogram recordings: The george loody PhysioNet challenge 2022. *PLOS Digital Health*. <https://doi.org/10.1371/journal.pdig.0000324>.
- B. Rocha, D. Pessoa, A. Marques, P. Carvalho, and R. P. Paiva. (2023). Automatic wheeze segmentation using harmonic-percussive source separation and empirical mode decomposition. *IEEE Journal of Biomedical and Health Informatics*. <https://doi.org/10.1109/jbhi.2023.3248265>.
- I. S. Roy, J. K. Roy, N. Mandal, and S. C. Mukhopadhyay. (2024). Recent advances in PCG signal analysis using AI: A review. *International Journal on Smart Sensing and Intelligent Systems*. <https://doi.org/10.2478/ijssis-2024-0012>.
- S. B. Shuvo, S. S. Alam, and T. Hasan. (2026). A self-attention-driven deep denoiser model for real time lung sound denoising in noisy environments. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2025.108818>.
- P. Somaratne, S. Herath, G. Gargiulo, P. Breen, N. Anderson, Y. Yao, T. Liu, and A. Withana. (2026). A dual classifier-regressor architecture for heart sound onset/offset detection. *IEEE transactions on bio-medical engineering*. <https://doi.org/10.1109/tbme.2026.3654558>.

- 9 W. Sun, Y. Zhang, and F. Chen. (2024). Research on heart and lung sound separation method based on DAE-NMF-VMD. *EURASIP Journal on Advances in Signal Processing*. <https://doi.org/10.1186/s13634-024-01152-0>.
- 1 Y. Torabi, S. Shirani, and J. P. Reilly. (2023). A new non-negative matrix factorization approach for blind source separation of cardiovascular and respiratory sound based on the periodicity of heart and lung function. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2305.01889>.
- 21 Y. Torabi, S. Shirani, and J. P. Reilly. (2025a). Descriptor: Heart and lung sounds dataset recorded from a clinical manikin using digital stethoscope (HLS-CMDS). *IEEE Data Descriptions*, 2, 133–140. <https://doi.org/10.1109/IEEEDATA.2025.3566012>.
- 1 Y. Torabi, S. Shirani, and J. P. Reilly. (2025b). Large language model-based nonnegative matrix factorization for cardiorespiratory sound separation. *arXiv.org*. <https://doi.org/10.48550/arxiv.2502.05757>.
- 37 I.-S. Tsai, C.-C. Hsu, C.-L. Hsieh, and Y. Liu. (2023). Separation of heart and lung sounds by a deep network-based model. <https://doi.org/10.1109/ickii58656.2023.10332660>.
- 46 R. Ullah and S. Zhang. (2024). Unsupervised framework for single channel heart and lung sounds separation in data constrained environments. *Applied Acoustics*. <https://doi.org/10.1016/j.apacoust.2024.110076>.
- 1 W. Wang, D. Qin, S. Wang, Y. Fang, and Y. Zheng. (2023a). A multi-channel UNet framework based on SNM-DCNN for robust heart-lung-sound separation. *Computers in Biology and Medicine*. <https://doi.org/10.1016/j.compbiomed.2023.107282>.
- 1 W. Wang, S. Wang, D. Qin, Y. Fang, and Y. Zheng. (2023b). Heart-lung sound separation by nonnegative matrix factorization and deep learning. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2022.104180>.
- 45 Y. Wang, H. Yang, Z. Gao, Z. Dai, K. Yu, T. Song, H. Yang, Y. Wang, and J. Zhang. (2025). A phase-enhanced neural network with dual-path transformer for single-channel chest sound separation. *IEEE journal of biomedical and health informatics*. <https://doi.org/10.1109/jbhi.2025.3631643>.
- 10 B. Zhang, J. Wang, Z. Ye, L. Zhou, D. Xiong, X. Wang, and L. Guo. (2025). JASSNet: Heart and lung sound separation network based on joint attention mechanism and semi-supervised learning. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2025.107525>.
- 4 L. Zhang, F. He, H. Tan, W. Wang, and S. Wang. (2026). An efficient lung sound separation algorithm based on GIHO-VMD. *Computer methods and programs in biomedicine*. <https://doi.org/10.1016/j.cmpb.2025.109124>.
- 18 Q. Zhao, S. Geng, B. Wang, Y. Sun, W. Nie, B. Bai, C. Yu, F. Zhang, G. Tang, D. Zhang, Y. Zhou, J. Liu, and S. Hong. (2024). Deep learning in heart sound analysis: From techniques to clinical applications. *Health Data Science*. <https://doi.org/10.34133/hds.0182>.

- ⁸⁰ J. Zheng, Y. Wang, J. Niu, Y. Shi, and F. Xie. ²⁰ (2025). Identifying the respiratory sound based on single-channel separation and hyperdimensional computing. *IEEE Sensors Journal*. <https://doi.org/10.1109/jsen.2025.3557909>.
- ¹⁶ R. Zheng and S. Sanei. (2024). Separation of cardiopulmonary sound signals for classification of respiratory diseases. *Asia pacific signal and information processing association annual summit and conference (APSIPA ASC)*. <https://doi.org/10.1109/APSIPAASC63619.2025.10848841>.

Appendix A: Gantt Chart

Planned FYP1 Report Schedule for the Project under Review. The schedule below shows the general timing of report production for the project under review. It follows the proposed **CPT6314 Teaching Plan T2610** and current report workflow, from FYP1 briefing to submission of report for interim assessment, followed by preparation of oral presentation.



Appendix B: FYP1 Meeting Logs

CPT6314 Project (FYP1) Meeting Log

Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	06/04/2026	Meeting No.:	1	Meeting Mode:	☐ In-Person ☐ Online
Project ID:	985	Project Type:	☐ Research-based ☐ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:	1221303805	Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE
 (Please write the details of the work done, after the last meeting). Tasks (tick):

☐ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Confirmed project title, scope, and objectives.
- Set up report repository and literature review folders.
- Prepared paper screening and PRISMA tracking files.

2. WORK TO BE DONE
 (Please write the details of the work to be done, before the next meeting). Tasks (tick):

☐ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Collect and organize related papers.
- Define inclusion and exclusion criteria.
- Start screening existing papers.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
 (Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solution)

Some papers focus on disease diagnosis instead of sound separation.
Solution: Limit the review to cardiopulmonary sound separation only.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

The student has completed the initial project setup phase, including: 1. Confirmation of project title, scope, and objectives. 2. Establishment of report repository and literature review organization. 3. Preparation of paper screening and PRISMA tracking files. 4. Identification of relevant keywords and search terms. 5. Beginning systematic screening of literature.

Student's Signature: 

Supervisor's Signature: 

Co-Supervisor's Signature (if applicable):

Company Supervisor's Signature (if applicable):

IMPORTANT NOTES TO STUDENTS:

- Items 1 - 3 are to be completed by the students prior to the meeting.
- Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with their FYP2 report. Minimum requirement is SIX fortnightly meetings with the supervisor. (Weeks 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student by the supervisor and for checking the attendance requirement of the student. The supervisor / co-supervisor / company supervisor can provide feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Students who fail to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

Prepared by: AL-SALOUL, ASHRAF ALI HUSSEIN

CPT6314 Project I

2610 Term

CPT6314 Project (FYP1) Meeting Log

Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	27/04/2026	Meeting No.:	2	Meeting Mode:	☐ In-Person ☐ Online
Project ID:	985	Project Type:	☐ Research-based ☐ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:	1221303805	Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE
 (Please write the details of the work done, after the last meeting). Tasks (tick):

☐ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Screened papers from 2022-2026.
- Recorded Include, Maybe, and Exclude decisions.
- Updated references, extraction matrix, and PRISMA counts.

2. WORK TO BE DONE
 (Please write the details of the work to be done, before the next meeting). Tasks (tick):

☐ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Download accessible papers using Zotero.
- Audit newly downloaded PDFs.
- Prepare Chapter 2 evidence map.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
 (Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solution)

Some papers were not accessible through university access.
Solution: Marked them as excluded and recorded the reasons for PRISMA.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

The student has completed the initial project setup phase, including: 1. Confirmation of project title, scope, and objectives. 2. Establishment of report repository and literature review organization. 3. Preparation of paper screening and PRISMA tracking files. 4. Identification of relevant keywords and search terms. 5. Beginning systematic screening of literature.

Student's Signature: 

Supervisor's Signature: 

Co-Supervisor's Signature (if applicable):

Company Supervisor's Signature (if applicable):

IMPORTANT NOTES TO STUDENTS:

- Items 1 - 3 are to be completed by the students prior to the meeting.
- Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with their FYP2 report. Minimum requirement is SIX fortnightly meetings with the supervisor. (Weeks 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student by the supervisor and for checking the attendance requirement of the student. The supervisor / co-supervisor / company supervisor can provide feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Students who fail to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



CPT6314 Project (FYP1) Meeting Log
Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	18/05/2026	Meeting No.:	3	Meeting Mode:	☐ Research-based	☐ In-Person	☐ Online
Project ID:	985	Project Type:	☐ Research-based	☐ Application-based			
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation						
Student ID:	1221303805	Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN				
Student Programme and Specialisation:	Bachelor of Computer Science (Hons.) / Software Engineering						
Supervisor Name:	Arshad Jamal						
Co-Supervisor Name: (if applicable)	N/A						
Collaborating Company: (if applicable)	N/A						
Company Supervisor Name: (if applicable)	N/A						

1. WORK DONE
(Please write the details of the work done, after the last meeting/ Tasks (tick):
☐ Background study ☐ Implementation ☐ Testing ☐ Evaluation
☐ Communication/ Proposal ☐ Research Paper ☐ Draft Report ☐ Final Report
Details (max 3-5 bullet points):

- Finalized Chapter 2 evidence map.
- Drafted Chapter 1, Chapter 2, and Chapter 3.
- Generated the report in Word format using Quarto.

2. WORK TO BE DONE

(Please write the details of the work to be done, before the next meeting/ Tasks (tick):
☐ Background study ☐ Implementation ☐ Testing ☐ Evaluation
☐ Communication/ Proposal ☐ Research Paper ☐ Draft Report ☐ Final Report
Details (max 3-5 bullet points):

- Add literature review matrix in Chapter 2.
- Number the problem statements and align them with objectives.
- Fix report formatting and front matter.
- Insert the final PRISMA diagram into Chapter 2.
- Start Chapter 4: system design and implementation.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

(Please write the details of the problems encountered, after the last meeting, and provide the solutions / plan for the solutions)

The report still needs clearer alignment between the problem statements and project objectives. Chapter 2 also needs a literature review matrix to summarize the reviewed papers clearly. Solution: Reverse Chapter 1 by numbering the problems and matching them with the objectives. Update Chapter 2 by adding the literature review matrix based on the selected papers.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

Align the problem statements and objectives. Integrate final Prisma diagram. Prepare a literature review paper in MDPI format.

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature (if applicable)

Company Supervisor's Signature (if applicable)

IMPORTANT NOTES TO STUDENTS:

- Items 1–3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX (6) FYP meeting logs (Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on their progress. It is the responsibility of the student to ensure that they provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- When all the meeting logs are submitted, the supervisor / co-supervisor / company supervisor will not be allowed to submit FYP papers.

Appendix C: Turnitin Similarity Index Page

ALSALOUL ASHRAF ALI HUSSEIN - 1221303805_Arshad_Jamal_FYP1_Report			
ORIGINALITY REPORT			
16 %	12 %	12 %	9 %
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS
PRIMARY SOURCES			
1 Submitted to Multimedia University			
Student Paper			
2 arxiv.org			
Internet Source			
3 sameni.info			
Internet Source			
4 documentsdelivered.com			
Internet Source			
5 Yang Yi Poh, Ethan Grooby, Kenneth Tan, Atul Malhotra, Mehrtash Harandi, Faezeh Marzbaniarad, "Reprogramming Automatic Speech Recognition Models for Neonatal Che Sound Separation", IEEE Journal of Biomedical and Health Informatics, 2026			
Publication			
6 Shams Nafisa Ali, Afia Zahin, Samiul Based Shuvo, Nusrat Binta Nizam et al. "BUET multi disease heart sound dataset: A comprehensive auscultation dataset for developing computer-aided diagnostic systems", Computer Methods and Programs in Biomedicine Update, 2026			
Publication			
7 research.monash.edu			
Internet Source			
8 ackerisim.harran.edu.tr:8080			
Internet Source			
9 digitalcommons.odu.edu			
Internet Source			
10 pubmed.ncbi.nlm.nih.gov			
Internet Source			
11 Leiming Zhang, Fullang He, Hao Tan, Wangquan Wang, Shiyuan Wang, "An efficient lung sound separation algorithm base on GHO-VMD", Computer Methods and Programs in Biomedicine, 2026			
Publication			
12 Submitted to Aalto Yliopisto			
Student Paper			

13 ouci.dntb.gov.ua			
Internet Source			
14 etd.repository.ugm.ac.id			
Internet Source			
15 Jaspreet Kaur, Brijesh Kumar, "Comparative Analysis of Machine Learning Algorithms for Lung Sound Analysis for Disease Prediction", 2025 3rd International Conference on Advances in Computation, Communication and Information Technology (ICACCIIT), 2026			
Publication			
16 Submitted to University of North Texas			
Student Paper			
17 vbn.aau.dk			
Internet Source			
18 Jing Lyu, Sijie Su, Xin Wang, Enhai Song, Qinyuan Zhang, " High Performance Flexible Optical Fiber Sensor Enabled by Er/Yb Co-Doped near Pure-Green Upconversion Emission Phosphor for Implantable Temperature Sensing ", Advanced Materials Interfaces, 2026			
Publication			
19 research.buaa.edu.cn			
Internet Source			
20 www.mdpi.com			
Internet Source			
21 data.mendeley.com			
Internet Source			
22 Wenhui Sun, Yipeng Zhang, Fuming Chen, "Research on heart and lung sound separation method based on DAE-NMF-VMD", EURASIP Journal on Advances in Signal Processing, 2024			
Publication			
23 Daisuke Niizumi, Daiki Takeuchi, Masahiro Yasuda, Binh Thien Nguyen, Yasunori Ohish Noboru Harada, "Assessing the Utility of Audio Foundation Models for Heart and Respiratory Sound Analysis", 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2025			
Publication			
24 Submitted to University of South Wales - Pontypridd and Cardiff			
Student Paper			
25 Maros Jakubec, Eva Lieskovska, Peter Pocka, "A robust deep learning based model for denoising phonocardiogram signals in clinical environments", Engineering Applications Artificial Intelligence, 2025			
Publication			

- 36 hdl.handle.net
Internet Source
- 37 Francis Chikweto, Muyanga Kampeketi, Mubanga Boyd, Kalupa Conix, Munsanje Foste Chilufya Queen. "Toward the Development of a Low-Cost AI-Enabled Digital Stethoscope 2024 International Conference on Electrical and Computer Engineering Researches (ICECER), 2024
Publication
- 38 Submitted to Nightingale College - School of Nursing
Student Paper
- 39 Peng-Fei Zhang, Hua-Nong Ting, Siow-Wee Chang. "Using Cough Sounds for the Recognition of Multiclass Respiratory Diseases With Artificial Intelligence: A Review", IET Access, 2025
Publication
- 40 Van-Thuan Tran, Ting-Hao You, Wei-Ho Tsai. "Acoustic Cues for Person Identification Using Cough Sounds", Computer Methods and Programs in Biomedicine Update, 2025
Publication
- 41 Alessandra Fava, Behnood Dianat, Alessandro Bertacchini, Andreina Manfredi, Marco Sebastiani, Marco Modena, Fabrizio Pancaldi. "Pre-processing techniques to enhance the classification of lung sounds based on deep learning", Biomedical Signal Processing and Control, 2024
Publication
- 42 Submitted to University Of Tasmania
Student Paper
- 43 bura.brunel.ac.uk
Internet Source
- 44 sciendo.com
Internet Source
- 45 Submitted to Nanyang Technological University
Student Paper
- 46 open.library.ubc.ca
Internet Source
- 47 Ming Hung Wong. "Handbook of Microplastics and Associated Chemicals - Ecological and Health Impacts", CRC Press, 2026
Publication
- 48 Xu Wang, Hui Zhou, Yanjie Xu, Yingtao Zhang, Zhenguo Nie. "Multimodal Acoustic Feature Fusion with Hybrid Deep Networks for Industrial Equipment Fault Detection", 2025 International Conference on Equipment Intelligent Operation and Maintenance (ICEION 2025)
Publication
- 39 nitrkl.ac.in
Internet Source
- 40 Chenwen Wu, Na Ye, Jialin Jiang. "Classification and Recognition of Lung Sounds Based on Improved Bi-ResNet Model", IEEE Access, 2024
Publication
- 41 doaj.org
Internet Source
- 42 Bowen Han, Wei Quan. "Cardiorespiratory Sound Separation Using Singular Spectrum Analysis", 2025 17th International Conference on Signal Processing Systems (ICSPS), 2025
Publication
- 43 Malaya Kumar Nath, Sanghamitra Subhadarsini Dash, Karthick Mookkandi, Thiya A.. "Towards intelligent phonocardiogram analysis: Advances and future perspectives in cardiovascular disease detection", Engineering Applications of Artificial Intelligence, 2025
Publication
- 44 Rizwan Ullah, Shaohui Zhang. "Unsupervised framework for single channel heart and lung sounds separation in data constrained environments", Applied Acoustics, 2024
Publication
- 45 iieta.org
Internet Source
- 46 Submitted to Faculty of Medicine & Health Sciences
Student Paper
- 47 Subhasikta Behera, Puneet Kumar Jain, Sambit Bakshi. "Adaptive thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds", Biomedical Signal Processing and Control, 2026
Publication
- 48 "Table of Contents", IEEE Journal of Biomedical and Health Informatics, 2026
Publication
- 49 Wangquan Wang, Chenghong Bo, An Zhao, Leiming Zhang, Fuliang He. "Heart-lung sound unsupervised separation by nonnegative matrix factorization and modified Hilbert spectrum", Biomedical Signal Processing and Control, 2025
Publication
- 50 assets-eu.researchsquare.com
Internet Source
- 51 www.newbornjournal.org
Internet Source
- 52 Submitted to University of Stellenbosch, South Africa
Student Paper

- 53 www.xiaolei-zhang.net
Internet Source
- 54 Bochao Zhang, Jiping Wang, Zhipeng Ye, Linfu Zhou, Daxi Xiong, Xiaojun Wang, Liqian Guo. "JASSNet: Heart and lung sound separation network based on joint attention mechanism and Semi-Supervised learning", Biomedical Signal Processing and Control, 2025
Publication
- 55 Submitted to University of Pretoria
Student Paper
- 56 Yanshan Sha, Dongliang Chen, Feiyang Wu, Qiang Cao, Yucong Zhou, Weihua Zhang, Xi Huang, Minghui Wang. "Experimental investigation of time-frequency characteristics of acoustic signals during boilover in small-scale oil tanks", Journal of Loss Prevention in the Process Industries, 2026
Publication
- 57 erepo.uef.fi
Internet Source
- 58 jeemil.org
Internet Source
- 59 suraj.lums.edu.pk
Internet Source
- 60 H.L. Gururaj, Francesco Flammini, S. Srividhya, M.L. Chayadevi, Sheba Selvam. "Computational Science Engineering", CRC Press, 2024
Publication
- 61 Yuqi Wang, Han Yang, Zhixing Gao, Zhiwei Dai, Kang Yu, Tingting Song, Hao Yang, Yunfei Wang, Jun Zhang. "A Phase-Enhanced Neural Network With Dual-Path Transformer for Single-Channel Chest Sound Separation", IEEE Journal of Biomedical and Health Informatics, 2026
Publication
- 62 digibuo.uniovi.es
Internet Source
- 63 ds.inflibnet.ac.in
Internet Source
- 64 "Swarm, Evolutionary, and Memetic Computing", Springer Science and Business Media LLC, 2015
Publication
- 65 Aymen Anwar, Yassin Khalifa, James L. Coyle, Ervin Sejdic. "Transformers in biosignal analysis: A review", Information Fusion, 2025
Publication
- 66 Ravi Pal, Anna Barney. "Automatic breathing phase identification based on the second derivative of the recorded lung sounds", Biomedical Signal Processing and Control, 2025
Publication
- 67 github.com
Internet Source
- 68 "2023 Index IEEE Journal of Biomedical and Health Informatics Vol. 27", IEEE Journal of Biomedical and Health Informatics, 2023
Publication
- 69 www.papers.phmsociety.org
Internet Source
- 70 "Applications of Computational Intelligence in Management & Mathematics", Springer Science and Business Media LLC, 2023
Publication
- 71 E. Grooby, C. Sitaula, D. Fattahi, R. Samei, K. Tan, L. Zhou, A. King, A. Ramanathan, A. Malhotra, G.A. Dumont, F. Marzbanrad. "Noisy Neonatal Chest Sound Separation for High Quality Heart and Lung Sounds", IEEE Journal of Biomedical and Health Informatics, 2025
Publication
- 72 researchr.org
Internet Source
- 73 www.peeref.com
Internet Source
- 74 Submitted to British University in Egypt
Student Paper
- 75 Lok, Richard Him. "Tool Support for Software Process Assessments", University of the Witwatersrand, Johannesburg (South Africa), 2025
Publication
- 76 WeiBo Wang, Shubo Wang, Dimei Qin, Yu Fang, Yongkang Zheng. "Heart-lung sound separation by nonnegative matrix factorization and deep learning", Biomedical Signal Processing and Control, 2023
Publication
- 77 Yasaman Torabi, Shahram Shirani, James P. Reilly. "LingoNMF: Large language model-enhanced nonnegative matrix factorization for cardiorespiratory sound separation", Biomedical Signal Processing and Control, 2026
Publication
- 78 assets.researchsquare.com
Internet Source
- 79 online-journals.org
Internet Source

80	portalcris.ismuni.it	Internet Source
81	scholarshare.temple.edu	Internet Source
82	Submitted to upGrad	Student Paper
83	Ethan Grooby, Jinyuan He, Julie Kiewsky, Davood Fattahi et al. "Neonatal Heart and Lung Sound Quality Assessment for Robust Heart and Breathing Rate Estimation for telehea Applications", IEEE Journal of Biomedical and Health Informatics, 2020	Publication
84	Submitted to Sunway Education Group	Student Paper
85	medinform.jmir.org	Internet Source
86	mobt3ath.com	Internet Source
87	Meltem Çınar Bozdağ, Alper Kumcu, Lütfi Kerem Şenel, Havva Nur Temizkan et al. "Spor language biomarkers in Turkish-speaking schizophrenia patients: Evidence from linguistic analysis and word embeddings", Psychiatry Research, 2026	Publication
88	Submitted to Sheffield Hallam University	Student Paper
89	ijmada.com	Internet Source
90	os.unilcloud.switch.ch	Internet Source
91	pdfs.semanticscholar.org	Internet Source
92	researchbank.swinburne.edu.au	Internet Source
93	scholar.hit.edu.cn	Internet Source
94	Arshad Jamal, R. Kanesaraj Ramasamy, Junaidi Abdullah, Sivasutha Thanjappan, Faizal Amri Hamzah, Shamsuriani Bt Md Jamal. "Chapter 4 Energy-Efficient Biomedical Signal	

Decoupling Via Optimized Generative AI Models", Springer Science and Business Media LLC, 2026		
95	Submitted to University of the West Indies	Student Paper
96	ase.in.tum.de	Internet Source
97	baltijapublishing.lv	Internet Source
98	cdn.techscience.cn	Internet Source
99	ebin.pub	Internet Source
100	researchbank.rmit.edu.au	Internet Source
101	www.coursehero.com	Internet Source
102	Ali Hussein Shaker, Ibrahim Amer Ibrahim, Sadik Kamel Gharghan. "Machine learning techniques for cardiovascular disease detection through heart sound analysis: A review AIP Publishing, 2024	Publication
103	Arvind Dagur, Dhirendra Kumar Shukla, Nazarov Fayzullo Makhmadiyrovich, Akhatov Akmal Rustamovich, Jabborov Jamol Sindorovich. "Artificial Intelligence and Information Technologies", CRC Press, 2024	Publication
104	Submitted to Capella University	Student Paper
105	Green, Virginia. "Entrepreneurial boot camps: A case study to evaluate and improve the establishment of methods, goals, strategies and tasks", Proquest, 20111109	Publication
106	K. Sathesh, P Kowsalya, E. Aravindraj, S.Guru Naveen Reddy, K. Anvith Sai Varma, P. Ganesh. "Clearer HeartBeats: Enhancement of cardiac sounds using Adaptive Filtering, Wavelet Decomposition", 2024 10th International Conference on Communication and Signal Processing (ICCSPP), 2024	Publication
107	Submitted to Midlands State University	Student Paper

109

R. N. V. Jagan Mohan, B. H. V. S. Rama Krishnam Raju, V. Chandra Sekhar, T. V. K. P. Pra: "Algorithms in Advanced Artificial Intelligence - Proceedings of International Conference on Algorithms in Advanced Artificial Intelligence (ICAAAI-2024)", CRC Press, 2025
Publication

109

Yuan Xie, Kan Xie, Qiyu Yang, Shengli Xie. "Reverberant blind separation of heart and lu sounds using nonnegative matrix factorization and auxiliary function technique", Biomedical Signal Processing and Control, 2021
Publication

110

eprints.bournemouth.ac.uk
Internet Source

111

eprints.utar.edu.my
Internet Source

112

es.scribd.com
Internet Source

113

etd.uum.edu.my
Internet Source

114

journal.50sea.com
Internet Source

115

journals.plos.org
Internet Source

116

knowledge.lancashire.ac.uk
Internet Source

117

par.nsf.gov
Internet Source

118

research.utwente.nl
Internet Source

119

studentsrepo.um.edu.my
Internet Source

120

www.csauthors.net
Internet Source

121

Adekanmbi, Motunrayo Helen. "Knowledge of Tuberculosis Patients About Their Diseases in Tshwane, South Africa", University of South Africa (South Africa)
Publication

122

Chunjian Yang, Nan Hu, Dongyang Xu, Zhi Wang, Shengsheng Cai. "Monaural cardiopulmonary sound separation via complex-valued deep autoencoder and cyclostationarity", Biomedical Physics & Engineering Express, 2023
Publication

123

Mahmoud Fakhry, Abeer FathAllah Brery. "Generative artificial intelligence for heart sound analysis: From model architectures to clinical applications", Intelligent Systems y Applications, 2026
Publication

124

Przemyslaw Pietrzak, Marcin Wolkiewicz. "Demagnetization Fault Diagnosis of Permanent Synchronous Motors Based on Stator Current Signal Processing and Machine Learning Algorithms", Sensors, 2023
Publication

125

Shuvashis Sarker, Faika Fairuj Preotee, Tashreef Muhammad, Shamim Akhter. "An Innovative Framework for Heart Sound Classification Integrating Adaptive Fuzzy Rank-Based Ensemble of Transfer Learning Models with Multi-Dimensional Features", Cold Spring Harbor Laboratory, 2025
Publication

126

Yin Liang, Xiaoying Yang, Runrui Han, Yingchen Jia, Gaoxu Xu. "A learnable multi-order spatial-temporal attention network for fMRI-based neurological disorder identification Biomedical Signal Processing and Control, 2026
Publication

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	Off		

Appendix D: Full Literature Review Matrix

The whole literature review matrix below provides an opportunity to assess the sources included in the chapter 2 synthesis. It allows for comparing how each of the cited works addressed different research aims, approaches, data sets, results, limitations, and applicability to the proposed cardiopulmonary sound separation system.

No.	Research Title	Author(s) & Year	Journal / Conference / Database	Research Problem / Objective	Algorithm / Framework Used	Dataset Used	Results / Findings	Research Gap	Limitation	How It Helps This Project
1	⁶ An End-to-End Deep Learning Framework for Real-Time Denoising of Heart Sounds for Cardiac Disease Detection in ⁶⁵Seen Noise	Ali et al. (2023) (Ali et al., 2023)	IEEE Access	Improve noisy heart sound recordings before later analysis.	LU-Net encoder-decoder denoiser.	PCG data mixed with breathing and hospital noise.	Reported SNR and real-time denoising improvements.	Denoising is related but does not separate heart and lung outputs.	Focuses on heart sound denoising and diagnosis support.	Supports preprocessing and noise-reduction choices.
2	⁶⁵ Noisy Neonatal Chest Sound Separation for High-Quality Heart and Lung Sounds	Grooby et al. (2023) (Grooby et al., 2023)	⁸¹ IEEE Journal of Biomedical and Health Informatics ⁴⁰	Separate noisy neonatal chest recordings into heart and lung sounds.	NMF and NIMCF.	Neonatal artificial mixtures and noisy chest sounds.	Improved signal quality and vital-sign estimation support.	Robust separation remains difficult under noisy neonatal conditions.	Dataset is neonatal and not clearly public.	Provides a classical separation baseline.
3	⁸ NeoSNet: Real-Time Neonatal Chest Sound Separation Using Deep Learning	Poh et al. (2024) (Poh et al., 2024)	IEEE Open Journal of Engineering in Medicine and Biology	Perform real-time neonatal chest sound separation.	1D encoder-decoder with transformer mask generation.	Neonatal chest recordings with heart and lung references.	Reported SDR and SI-SDR improvement with real-time relevance.	Reproducibility is limited by private clinical data.	Performance depends on noise and recording quality.	Main deep learning reference for the prototype.
4	³² Recent Advances in PCG Signal Analysis using AI: A Review	Roy et al. (2024) (R ³² et al., 2024)	International Journal on Smart Sensing and Intelligent Systems	Review AI methods for PCG preprocessing and analysis.	Review of preprocessing, segmentation, features, and AI models.	PCG datasets from prior studies.	Summarizes common PCG processing stages and AI approaches.	Review is not focused on heart-lung source separation.	Mainly classification and PCG analysis oriented.	Supports background on PCG preprocessing.
5	²² Research on Heart and Lung Sound Separation	Sun et al. (2024) (Sun et al., 2024)	EURASIP Journal on Advances in	In ²² ve heart-lung sound separation using	Deep autoencoder, NMF, and VMD.	Heart-lung sound mixtures.	Reported SDR, PESQ, STOI, and	Reproducible public data access is limited.	Data and code access may require request.	Shows how ML and decomposition

	Method Based on		Signal Processing	a hybrid method.			separation metrics.		can be combined.
6	⁴² Unsupervised Framework for Single Channel Heart and Lung Sounds Separation in Data Constrained Environments	Ullah and Zhang (2024) (Ullah and Zhang, 2024)	Applied Acoustics	⁴² Separate single-channel heart and lung sounds with limited data.	MCRA, OMLSA, and NMF-HMM.	Heart and lung sound mixtures.	Reported SDR, SAR, and separation performance.	Data-constrained separation needs methods with limited labels.	Supports feasible low-data separation options.
7	¹² Multi-Channel UNet Framework Based on SNMF-DCNN for Robust Heart-Lung-Solids Separation	Wang et al. (2023) (Wang et al., 2023a)	Computers in Biology and Medicine	Improve robust heart-lung sound separation.	SNMF-DCNN-MCUNet.	Heart-lung mixtures with different lung sound types.	⁷⁴ Reported SNR, SDR, SIR, SAR, and correlation improvements.	Constructed mixtures affect generalizability.	Guides U-Net-style model design and metrics.
8	⁷¹ Sound Separation by Nonnegative Matrix Factorization and Deep Learning	Wang et al. (2023) (Wang et al., 2023b)	Biomedical Signal Processing and Control	Combine NMF and deep learning for separation.	MCNMF, clustering, and ECNet.	Heart-lung sound mixtures.	⁷⁴ Reported SNR, SDR, SIR, SAR, and correlation.	Separation depends on mixture assumptions.	Supports hybrid baseline and evaluation planning.
⁵⁰ 9	⁵¹ JASSNet: Heart and Lung Sound Separation Network Based on Joint Attention Mechanism and Semi-Supervised Learning	Zhang et al. (2025) (Zhang et al., 2025)	Biomedical Signal Processing and Control	Improve separation using attention and semi-supervision.	Joint attention and semi-supervised network.	Simulated and clinical cardiopulmonary recordings.	Reported SDR, SIR, SAR, and rate-estimation improvement.	Advanced models can be difficult to reproduce.	Defines a recent advanced comparison method.
10	⁷⁸ Deep Learning in Heart Sound Analysis: From Techniques to Clinical Applications	Zhao et al. (2024) (Zhao et al., 2024)	Health Data Science	¹²³ Review deep learning for heart sound analysis.	Review of CNN, RNN, transformer, denoising, and segmentation methods.	Common heart sound datasets.	Summarizes heart sound AI methods and dataset issues.	Not focused on heart-lung source separation.	Supports ML background and dataset discussion.
11	¹⁶ Separation of Cardiopulmonary Sound Signals for Classification of	Zheng and Sanei (2024) (Zheng and Sanei, 2024)	APSIPA ASC	Separate cardiopulmonary signals before	Singular spectrum analysis with	Auscultated cardiopulmonary recordings.	Separation improved downstream	Separation quality is secondary to classification.	Shows separation value without shifting project

	Respiratory Diseases			respiratory analysis.	kurtosis optimization.		respiratory classification.		scope to diagnosis.
12	8 Reprogramming Automatic Speech Recognition Models for Neonatal Chest Sound Separation	Poh et al. (2026) (Poh et al., 2026)	7 IEEE Journal of Biomedical and Health Informatics	Adapt speech models for neonatal chest sound separation.	Whisper model reprogramming.	Neonatal chest mixtures and supporting PCG data.	Reported separation, segmentation, and rate-estimation results.	Foundation-model adaptation is complex.	Identifies an advanced ML direction beyond the first prototype.
13	Enhanced Neural Network With Dual-Path Transformer for Single-Channel Chest Sound Separation	34 Wang et al. (2025) (Wang et al., 2025)	42 IEEE Journal of Biomedical and Health Informatics	Improve single-channel chest sound reconstruction.	Phase-enhanced dual-path transformer.	Constructed heart and lung sound mixtures.	Reported SDR-based and phase-aware improvements.	Phase reconstruction is often under-addressed.	Supports time-frequency and reconstruction design choices.
14	1 Large Language Model-based Nonnegative Matrix Factorization For Cardiorespiratory Sound Separation	Torabi et al. (2025) (Torabi et al., 2025b)	arXiv	Use language-model feedback to tune NMF separation.	LingoNMF with LLaMA-based feedback.	Synthetic and manikin heart-lung recordings.	Reported SNR, SDR, SIR, and SAR improvements.	LLM-assisted separation adds system complexity.	Helps compare explainable NMF extensions.
15	A New Non-Negative Matrix Factorization Approach for Blind Source Separation of Cardiovascular and Respiratory Sound Based on the Periodicity of Heart and Lung Function	Torabi et al. (2023) (Torabi et al., 2023)	arXiv	Use periodicity to separate cardiovascular and respiratory sounds.	Periodicity-guided NMF.	Synthetic heart and lung sound mixtures.	Reported SDR, SIR, and SAR.	Real-world validation remains limited.	Supports physiological periodicity as a separation cue.
16	116 Separation of Heart and Lung Sounds by A	Tsai et al. (2023) (Tsai et al., 2023)	International Conference on Knowledge	Separate generated heart-lung mixtures	Deep network-based model.	Public heart and lung sounds used to	Reported correlation-based	Evaluation needs stronger metrics and real recordings.	Provides a feasible student-level ML reference.

	Deep Network-Based Model		Innovation and Invention	using a deep model.		generate mixtures.	separation results.			
17	¹³ An efficient lung sound separation algorithm based on GIHO-VMD	Zhang et al. (2025) (Zhang et al., 2026)	Computer Methods and Programs in Biomedicine	⁶⁸ Separate lung sounds from mixed heart-lung signals.	GIHO-VMD.	⁶⁸ e-simulated mixed heart and lung sound signals.	Reported SNR, NCC, spectrogram, and runtime comparisons.	Decomposition methods remain sensitive to simulation assumptions.	Real-world application pipeline is not fully shown.	Provides a recent decomposition comparator.
18	²⁰ Identifying the Respiratory Sound Based on Single-Channel Separation and Hyperdimensional	³⁴ Zheng et al. (2025) (Zheng et al., 2025)	IEEE Sensors Journal	Use separation before respiratory sound identification.	Encoder-decoder separation, hyperdimensional computing, and LightGBM.	ICU respiratory recordings and benchmark data.	Reported respiratory identification improvements.	Separation is used mainly for classification.	Standalone separation quality is not the main objective.	Shows how separation can support later audio analysis.
19	⁵⁴ Computing Descriptor: Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope (H12-CMDS) Automatic Wheeze Segmentation Using Harmonic-Percussive Source Separation and Empirical Mode Decomposition	Torabi et al. (2025) (Torabi et al., 2025a)	IEEE Data Descriptions	Provide paired recordings for separation research.	Dataset descriptor.	HLS-CMDS manikin heart, lung, mixed, and source recordings.	Provides mixed recordings with corresponding source signals.	Public paired source datasets remain limited.	Manikin data may not represent full patient variability.	Main public dataset candidate for training and evaluation.
20	¹⁴ Wheeze Segmentation Using Harmonic-Percussive Source Separation and Empirical Mode Decomposition	Rocha et al. (2023) (Rocha et al., 2023)	IEEE Journal of Biomedical and Health Informatics	Segment wheezes from respiratory recordings.	HPSS and EMD.	Respiratory sound datasets.	Reported wheeze segmentation performance.	Respiratory preprocessing is not heart-lung separation.	Full real-world wheeze segmentation remains difficult.	Supports preprocessing and respiratory signal handling.
21	³³ A Blind Filtering Framework for Noisy Neonatal Chest Sounds	Fattahi et al. (2022) (Fattahi et al., 2022)	IEEE Access	Filter noisy neonatal chest sounds.	Wavelet transform, PCA, periodic component analysis, and SOBI.	Neonatal chest sound recordings.	Improved quality assessment and rate estimation.	Traditional filtering needs careful component selection.	Severe artefacts may still reduce reliability.	Supports preprocessing and baseline separation design.

22	A Dual Classifier-Regressor Architecture for Heart Sound Onset/Offset Detection	Somarathne et al. (2026) (Somarathne et al., 2026)	IEEE Transactions on Biomedical Engineering	Detect heart sound onset and offset.	Classifier-regressor PCG architecture.	PhysioNet/CinC heart sound data.	Reported timing and segmentation metrics.	Segmentation is not source separation.	Requires PCG/ECG-style context.	Supports timing and evaluation considerations.
23	BUET multi-disease heart sound dataset: A comprehensive auscultation dataset for developing computer-aided diagnostic systems	Ali et al. (2026) (Ali et al., 2026)	Computer Methods and Programs in Biomedicine Update	Provide a public heart sound dataset.	Dataset descriptor.	BUET multi-disease heart sound recordings.	Provides labelled heart sound data for AI research.	Dataset is diagnosis-oriented, not paired separation data.	No lung source or mixed-source separation target.	Helps distinguish background datasets from separation datasets.
24	Cardiorespiratory Sound Separation Using Singular Spectrum Analysis	Han and Quan (2025) (Han and Quan, 2025)	41 International Conference on Signal Processing Systems	Separate cardiac and respiratory components.	Multi-stage singular spectrum analysis.	Digital stethoscope cardiorespiratory recordings.	Presents a recent direct separation approach.	Detailed evaluation evidence remains limited at this reporting stage.	Full implementation details need further confirmation.	Adds a recent non-deep-learning comparator.
25	SteinAid: A Digital Auscultation Platform for Pediatrics	Arjouni et al. (2023) (Arjouni et al., 2023)	Sensors	Support pediatric digital auscultation workflow.	Digital auscultation platform.	Pediatric auscultation use context.	Demonstrates platform-oriented auscultation support.	Platform design is not separation-specific.	Does not provide heart-lung separation output.	Supports software workflow and interface thinking.
26	Reliable Sound Dataset and Baseline Artificial Intelligence System for Respiratory Illnesses	Hoang et al. (2023) (Hoang et al., 2023)	PHM Society Asia-Pacific Conference	Provide respiratory audio dataset and baseline AI.	Dataset and baseline AI system.	Sound-Dr respiratory recordings.	Supports respiratory sound dataset context.	Respiratory illness focus differs from source separation.	No paired heart-lung source targets.	Provides supplementary dataset background.
27	Compressed Sensing of Acoustic Cardiorespiratory Signals Using a	Baeyens et al. (2025) (Baeyens et al., 2025)	IEEE EMBC	Reconstruct acoustic cardiorespiratory signals.	Compressed sensing with CNN reconstruction.	Acoustic cardiorespiratory signals.	Shows reconstruction quality can affect later processing.	Reconstruction is not heart-lung source separation.	Dataset and deployment details need confirmation.	Supports acquisition and reconstruction context.

	sound denoising in noisy environments						denoising potential.			
33	thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds	Biomedical Signal Processing and Control	Denoise real-life respiratory recordings.	DWT coefficient thresholding with UNet	Real-life respiratory sounds.	Supports multiscale denoising for noisy respiratory data.	Focuses on respiratory denoising rather than source separation.	No paired heart-lung outputs.	Supports wavelet and UNet preprocessing ideas.	
34	Lung sound signal denoising using discrete wavelet transform and artificial neural network	Biomedical Signal Processing and Control	Improve noisy lung sound quality.	DWT and artificial neural network.	Lung sound signals.	Supports classical and ML-based lung denoising.	Does not address cardiopulmonary source separation.	Dataset details need confirmation.	Supports preprocessing and denoising background.	
35	Automatic breathing phase identification based on the second derivative of the recorded lung sounds	Biomedical Signal Processing and Control	Identify breathing phases from lung sounds.	Second-derivative based phase identification.	Heterogeneous lung sound recordings.	Reported breathing phase identification metrics.	Phase identification is not source separation.	Focuses only on respiratory signal structure.	Supports lung sound segmentation and evaluation context.	

Appendix E: PRISMA Screening Summary

Below are the details of the PRISMA screening process used in Chapter 2.

Screening item	Count	Report interpretation
Records considered	96	Screened records relevant to cardiopulmonary sound separation and supporting literature
Records excluded	47	Excluded due to being out of scope, unavailable for supporting the current work, duplicates, or being irrelevant to the project
Eligible records after screening	49	Records left after the screening process and availability confirmation
Studies selected for Chapter 2 synthesis	35	Studies included in Chapter 2 synthesis, narrative, summary matrix, or full matrix
Core separation studies	18	Studies addressing cardiopulmonary, heart-lung, or chest separation
Background/supplementary studies	17	Studies addressing datasets, preprocessing, machine learning, evaluation, or other relevant topics
Eligible but unselected records	14	Records left as backup that were not used as evidence in the current paper

Appendix F: System Design Diagrams

The primary system design diagrams are presented in chapter 4. They include the prototype's context, use cases, activity, planned class structure, component-level architecture, and audio processing sequence. The diagrams are included to illustrate the design behind the prototype described in this thesis. All details not included in the diagrams were omitted for the sake of brevity.

Diagram	Purpose
Context diagram	The context diagram displays the relationships between the user, public repositories, and the proposed prototype.
Use case diagram	The use case diagram displays the cardiopulmonary sound separation system's use cases.
Activity diagram	The activity diagram displays the proposed system's sequence of activities from the request upload to download.
Class diagram	The class diagram displays the prototype's class structure and responsibilities.
Component diagram	The component diagram displays the browser interface, back-end API, service layer, model registry, separation strategies, database, and storage components.
Sequence diagram	The sequence diagram displays the processing sequence when the user uploads an audio file.

Appendix G: Planned Test Cases and Evaluation Metrics

The following planned tests are informed by the FYP2 implementation and evaluation. However, please note that only by implementing the prototype and conducting the tests, it will be possible to obtain results. In the table below, the test areas and the evidence for test completion are presented.

Test area	Planned check	Expected evidence
Audio upload	Accept a valid cardiopulmonary audio	Successful upload status and stored input metadata
File validation	Reject unsupported file type, missing file, or oversized file	Clear validation message not proceeding to the next step
Preprocessing	Convert audio into target sampling rate, channel format, and segment length	Stored preprocessing metadata and consistent model input
Model inference	Process a valid input of mixed audio with the selected method of separation	Heart and lung outputs
Output review	Provide the option to the user to interact, view, or download the separated outputs	Accessible final result view and downloadable .wav files
Metric calculation	Calculating evaluation metrics if reference sources are provided	An evaluation table with valid metric results
Error handling	Process a corrupted or incorrectly provided audio input without failure	User-informative error message and no erroneous output records

The table below includes the evaluation metrics that could be computed for cardiopulmonary prototypes if references sources were available.

Evaluation area	Planned metric or verification type	Purpose
Source separation quality	SDR, SI-SDR, SIR, or SAR if reference sources are available.	Estimate the degree of separation quality in relation to the target source.
Noise reduction	SNR or other signal quality indicator	Confirm that the preprocessing effectively reduced the level of noise
Audio reconstruction	Correlation, waveform inspection, or spectrogram comparison	Verify that the signal waveform was correctly reconstructed
System usability	Task completion and user feedback if user testing is approved.	Confirm that the system can be used to upload, process, and download the separated audio files.
Processing reliability	Planned test cases with success and failure conditions.	Verify that the prototype can consistently process both valid and invalid user inputs.

Appendix H: Full User Requirements Questionnaire

The following full set of questions was developed to establish traceability between the Google Form questions and the requirements analysis in chapter 3. In particular, although the requirements analysis in chapter 3 grouped the items under 10 broad requirement-focused categories for analysing and discussing the requirements, this appendix provides the full set of 24 questions as actually used in the Google Form survey for collecting the user requirements.



Machine Learning-Based System for Cardiopulmonary Sound Separation

User Requirements Questionnaire for Machine Learning-Based Cardiopulmonary Sound Separation System

Welcome and thank you for participating in this questionnaire.

This questionnaire is conducted as part of a Final Year Project at **Multimedia University (MMU)** involving the development of a machine learning-based system for cardiopulmonary sound separation. The proposed system aims to allow users to upload a mixed cardiopulmonary audio recording and obtain separated heart and lung sound outputs.

The purpose of this questionnaire is to understand user background, current challenges in working with biomedical or audio-processing tasks, and user preferences regarding system features, interface design, output presentation, and evaluation needs.

Your responses will help identify user requirements and support the design of the proposed prototype. There are no right or wrong answers. All responses will be used for academic purposes only and will remain confidential.

Estimated completion time: 3-5 minutes.

Section A - Respondent Background:

Tell us about your background and experience with audio, machine learning, or biomedical signals.

1. What is your age group? *

- ☐ Under 18
- ☐ 18-25
- ☐ 26-35
- ☐ 36 and above

2. What best describes you? *

- ☐ Software Engineering / Computer Science student
- ☐ AI / Machine Learning student
- ☐ Biomedical Engineering/ Health-related student
- ☐ Researcher
- ☐ Lecturer / Supervisor
- ☐ Other: _____

3. How familiar are you with audio processing or signal processing? *

- ☐ Not familiar
- ☐ Basic
- ☐ Intermediate
- ☐ Advanced

4. Have you worked with biomedical audio before? *

- ☐ Yes
- ☐ No
- ☐ Not sure

5. Which type of audio-related task are you most familiar with? *

- ☐ Audio recording
- ☐ Audio editing
- ☐ Audio classification
- ☐ Audio separation
- ☐ Machine learning model testing
- ☐ I have not worked with audio tasks before

Section B - Current Challenges in Cardiopulmonary AudioProcessing:

Help us understand the main problems in separating heart and lung sounds from mixed audio.

1. How difficult do you think it is to separate heart and lung sounds from a mixed cardiopulmonary recording? *

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

2. What problems do you think affect cardiopulmonary sound processing the most? *

- ☐ Background noise
- ☐ Overlap between heart and lung sounds
- ☐ Low recording quality
- ☐ Difficulty identifying useful sound components
- ☐ Lack of simple software tools
- ☐ Lack of clear output visualization
- ☐ Other: _____

3. Do you think noisy audio recordings can reduce the quality of machine learning-based sound processing? *

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

4. Do you think a simple software prototype for heart-lung sound separation would be useful for students or researchers? *

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

Section C - System Feature Preferences:

Choose the features you think would make this sound separation system more useful.

1. Which feature would be most important in the proposed system? *

- ☐ Upload mixed cardiopulmonary audio
- ☐ Preprocess noisy audio
- ☐ Separate heart and lung sounds
- ☐ Preview separated audio
- ☐ Download separated outputs
- ☐ View processing history
- ☐ View evaluation results

2. Which output would you prefer after sound separation? *

- ☐ Heart and lung audio files only
- ☐ Audio files with waveform visualization
- ☐ Audio files with spectrogram visualization
- ☐ Audio files with evaluation metrics
- ☐ All of the above

3. How important is it for the system to provide a preview of the separated heart and lung sounds? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

4. How important is it for the system to allow users to download separated audio outputs? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

5. How important is it for the system to show basic processing information, such as file name, processing status, and completion time? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

6. Should the system include a history page showing previously uploaded files and separation results? *

- ☐ Yes
- ☐ No
- ☐ Not sure

Section D - Interface and Output Preferences:

Tell us how you want the system to show results, audio previews, charts, and messages.

1. How would you prefer the separated results to be displayed? *

- ☐ Simple audio player only
- ☐ Audio player with waveform
- ☐ Audio player with spectrogram
- ☐ Audio player with waveform, spectrogram, and metrics
- ☐ Not sure

2. Which interface style would you prefer? *

- ☐ Simple interface with only main features
- ☐ Detailed interface with many technical options
- ☐ Balanced interface with simple view and optional advanced details

3. How important is it for the system interface to be easy for non-expert users? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

4. What information should appear first after separation is completed? *

- ☐ Download buttons for heart and lung sounds
- ☐ Audio preview players
- ☐ Waveform or spectrogram visualization
- ☐ Evaluation metrics
- ☐ Processing status summary

5. What type of error message would be most helpful if the uploaded file is invalid? *

- ☐ Short message only
- ☐ Message explaining the problem
- ☐ Message explaining the problem and suggested fix
- ☐ Not sure

Section E - Evaluation and Improvement:

Share what would make the system more accurate, useful, and easier to improve.

1. How important is it for the system to show evaluation metrics for the separated output? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

2. Which evaluation information would be most useful? *

- ☐ Signal quality score
- ☐ Separation accuracy score
- ☐ Noise reduction information
- ☐ Processing time
- ☐ All of the above

3. Would you use a prototype that separates mixed cardiopulmonary audio into heart and lung sound outputs? *

- ☐ Yes
- ☐ No
- ☐ Maybe

4. What improvement or feature would you like to see in a cardiopulmonary sound separation system?

Long answer text